On the identity of *-rare* in Passives and Potentials in Japanese *

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1 Introduction

As is well known, many languages use the same morpheme in a range of different constructions. In Russian, Turkish, Tetelcingo Nahuatl, for example, a potential reading (i.e. 'ability' or 'possibility') is expressed by the same morpheme that is found in a middle, reflexive, or passive construction (see Shibatani 1985, Kazenin 2001). This is also the case in Japanese, where the 'passive' suffix *-rare* occurs in passive and potential constructions−our focus in this paper−as well as in middle/spontaneous and subject honorific constructions (but not in reflexives). Why do we find essentially the same form in these constructions? Japanese linguists generally assume that this is a case of **accidental** homophony in contemporary Japanese, resulting from a historical development where the middle or passive morpheme 'grammaticalized' as a (potential) modal. The view that there are two distinct morphemes seems reasonable as potential *-(rar)e* differs not only in meaning but also (slightly) in form from passive *-(r)are*. Furthermore, potential *-rare* occurs prominently in active potentials. These have active alignment, as shown in (1) and (2).

Active Potentials:[1]

(1) Mei-ga kinoko-o tabe-rare -ru.
Mei-NOM mushroom-ACC eat-RARE -PRS
'Mei can eat mushrooms.'

(2) Ken-ga hayaku hasir -e -ru.
Ken-NOM quickly run -(RAR)E -PRS
'Ken can run quickly.'

In this paper, we depart from this view, and pursue the hypothesis that the homophony of passive and potential *-rare* is **non-accidental** even in contemporary Japanese: i.e. the same *-rare* occurs in the potential and passive. This is expected if there is a UG principle of **Anti-homophony** that rules out accidental homophony of functional categories (cf. Johns 1992, Embick 2003, Bobaljik 2012, Kayne 2017a, 2025, among others).[2]

Besides active potentials, however, there are potentials with passive alignment in Japanese (henceforth **psv-potentials**) (Teramura 1982a), as in the theme psv-potential in (3):

Psv-potentials: Theme passives

(3) Kinoko -ga (Mei -ni) tabe -rare -ru.
mushroom -NOM (Mei -DAT) eat -RARE -PRS
'Mushrooms are edible (by Mei).' ≈ 'Mushrooms can be eaten (by Mei).' (MOD>Voice$_{passive}$)

Psv-potentials play a central role in our paper. The Japanese literature generally does not acknowledge their passive properties, but largely follows Kuroda (1992, ch.8), who compares potentials with complex adjectival (i.e. tough movement) constructions, and casts the analysis in terms of Case alternations/'subjectivization' (see also Nakamura & Fujita 1998). Since Chomsky (1981) the general syntax literature appears to follow his (notoriously problematic) tough/A' movement analysis for psv-potentials (see Inoue 2004), and focuses on accusative/nominative alternations in active potentials (see fn 21 and §8.3).

Putting the principle of Anti-homophony to test on passive and potential *-rare*, however, implies that

[1] Abbreviations used in the glosses are: ACC(=Accusative), DAT (=Dative), GEN (=Genitive), NOM (=Nominative), TOP (=Topic), PRS(=Present Tense), PAST(=Past tense), PERF/PROGR or ASP(=Perfect/progressive *tei*), (R)ARE (=Passive), (RAR)E (=Potential).

[2] Kanno (1992) pursues a unified syntactic account of *-rare*. However, her account, as she acknowledges, fails to account for potential constructions (p.187). She pursues the idea that syntactic downgrading of the external argument (a core property of *-rare* constructions, also argued by Shibatani (1985) as a unifying property of all *rare* constructions), may be achieved in various ways, including lexical operations on the thematic grid (p.200). We exclude such accounts on theoretical grounds.

we must explore the hypothesis that the same passive morpheme *-rare* occurs in both constructions, and show how the passive and potential properties emerge from the anaysis.[3] Thus, similarities and differences in form, meaning, and syntactic properties of the two *-rare* constructions must be established and analyzed. To do so, we first establish the general landscape surrounding passives and psv-potentials, and extract the properties of (one) *-rare*. In fact, a substantial part of the paper is devoted to establishing and analyzing the fine structural hierarchies that are involved. Specifically, we propose that passive Voice *-rare* is a (sub)part of the potential. This allows the similarities and differences between the two *-rare* constructions to fall out in a modular fashion from (i) the properties of *-rare*, and (ii) the interaction of a single *-rare* with other elements in its local environment. The analysis we propose is guided by a set of theoretical assumptions, which fall within the boundaries of very general and widely accepted UG principles, as well as factors that have been identified as accounting for differences in constructions within a language. Here the possibility that a same item may be merged at different structural heights will be an important factor. Height of merge seems to be the standard account for the difference between, say, adjectival and verbal passives, or 'lexical' and 'syntactic' causatives in non-lexicalist accounts (Embick 2004, Pylkkänen 2008, and others: see §1.2 for further discussion).

In this paper, we adopt the (pre-theoretical) definition of the basic (canonical) passive given in the typology of Keenan & Dryer (2007): (i) **no agent** phrase (e.g, *by Mary*) is present; (ii) the main verb in its non-passive form is transitive; and (iii) the main verb expresses an action, taking (volitional) **agent** subjects and **patient** objects.[4] Importantly, we do not start out by taking a stand on whether the external argument is syntactically projected in the passive or not. We will let this emerge from the data and analysis.

1.1 Potential constructions: a first introduction

We first focus on passive potentials (**psv-potentials**), and do not discuss active potentials until §9. As seen in (4), Japanese potential constructions not only have a 'modal' component, but can also have a 'passive' component (cf. Teramura 1982: 259, Kanno 1992):

(4) **Psv-potentials**: Theme passives

a. Kinoko -ga (Mei -ni) tabe -rare -ru.
mushroom -NOM (Mei -DAT) eat -RARE -PRS
'Mushrooms are edible (by Mei).' ≈ 'Mushrooms can be eaten (by Mei).' (MOD>Voice$_{passive}$)

b. Sono hon -ga (Ken -ni) yom -e -ru.
that book -NOM (Ken -DAT) read -(RAR)E -PRS
'That book is readable (by Ken).' ≈ 'That book can be read (by Ken).' (MOD>Voice$_{passive}$)

The form of the potential suffix *-(rar)e* in standard contemporary Japanese is (nearly) identical to the passive suffix *-(r)are*. The form of the potential depends on whether the stem ends in a vowel or a consonant, i.e. on the phonological properties of the stem (see §3.1 for further discussion and references). For this reason, after vowel final stems, *-rare* is ambiguous between a psv-potential and a passive (4-a), but after consonant

[3] Nakamura & Fujita (1998) develop an analysis based on 'subjectivization'/case alternations, discussing alternations between obliques and nominatives. We return in fn (i) for an evaluation of their argument that potential *-(rar)e* cannot be the passive *-rare*.

[4] In their typological study, Keenan & Dryer (2007) establish that *by*-phases are not a necessary property of passives, but silent agents are, from G(eneralization)-1 *Some languages have no passive constructions* and G-2 *If a language has any passives it has ones characterized as basic*. G1 and G2 entail that if a language has basic passives with agent phrases then it has them without agent phrases. In other words *by*-phases are not a necessary property of passives, but silent agents are.

final stems, *-e* is unambiguous and only allows a psv-potential reading, as in (5-b).

(5) **Psv-potentials vs. Passives**

a. Kono doresu -ga (Mei -ni) ki -rare -ru. [Ambiguous]
this dress -NOM (Mei -DAT) wear -RARE -PRS
✓ 'This dress is wearable (by Mei).' ≈ 'This dress can be worn (by Mei).' (MOD>Voice$_{passive}$)
✓ 'This dress {is/will be} worn (by Mei).' (Voice$_{passive}$)

b. Sono kuruma -ga (Ken -ni) kaw -e -ru. [Unambiguous]
that car -NOM (Ken -DAT) buy -(RAR)E -PRS
✓ 'That car is buyable (by Ken).' ≈ 'That car can be bought (by Ken).' (MOD>Voice$_{passive}$)
NOT ok *'That car {is/will be} bought (by Ken).' (*Voice$_{passive}$)

Not only is the form in psv-potentials (nearly) identical to passive Voice *-rare*, it also shares core syntactic properties: promotion of some argument (or adjunct) to the nominative subject position, and 'absence' or 'silence' of the external argument. This kind of systematic study has not been undertaken before.[5] Psv-potentials, not only have a passive (sub)part, but also contain a (silent) modal component which we argue comes into two flavors, *ability* and *possibility*, corresponding to the 'root' (control-like) modality and a much higher 'alethic' modality respectively (i.e, Mod$_{possibility}$ >. . . Mod$_{ability}$) in Cinque's clausal hierarchy.[6] These high and low modality readings turn out to correlate with an important syntactic difference w.r.t. the type of *oblique passives/pseudopassives* (*ga*-marked obliques) that are allowed in passives and "low" *ability* psv-potentials, on the one hand, and in "high" *possibility* psv-potentials, on the other.

Besides canonical theme passives (aka direct passives), Japanese allows oblique passives/pseudopassives, as Ishizuka (2010, 2012) shows. These are generally called 'indirect passives' or 'adversity/gapless passives' in the literature. In oblique passives, a nominative DP alternates with an otherwise full PP.[7] We document that the same kind of oblique (and theme) passives allowed in the passive are possible in low psv-potentials, but that a wider variety of oblique passives can be found in high psv-potentials. In particular, goal (IO), directional 'to,' and source 'from' passives are possible in both low psv-potentials and passives. High psv-potentials additionally allow instrumental and locative (adjunct) passives.

(6) **Locative and instrumental high psv-potentials**[8] [Not ambiguous]

a. Kono kafe-ga oisii keeki-o tabe-rare -ru.
this cafe-NOM tasty cake-ACC eat-RARE -PRS (MOD>Voice$_{passive}$)
✓ 'This cafe can be eaten tasty cake in.' ≈'It is possible to eat tasty cake in this cafe.'
NOT ok *'This cafe {is/will be} eaten tasty cake in.' (* Voice$_{passive}$)

b. Sono kagi-ga kono doa-o ake -rare -ru.
that key-NOM this door-ACC open -RARE -PRS (MOD>Voice$_{passive}$)
✓'That key can be opened this door with.' ≈'It is possible to open this door with that key.'

[5] This might be due to the basic assumption the passive and potential are synchronically unrelated, thus raising the issue how potential constructions should be analyzed. Kuroda (1992, ch.8) shows (partial) similarities between (psv)-potentials and tough-movement constructions, but see Inoue (2004, ch.8) and Saito (1982). We briefly revisit the analysis of tough-movement in §10.3, from the perspective of our passive (i.e. A-movement) account, reviving earlier accounts of tough-movement.

[6] See Fukuda (2025) for a similar distinction between the two types of active potential modality.

[7] Goal and directional nominatives alternate with full PPs marked with *-ni*, source nominatives with *-kara*, and locative and instrumental nominatives with *-de*.

NOT ok *'That key {is/will be} opened this door with.' (*Voice$_{passive}$)

As shown in (6), locative and instrumental passives are always unambiguously (high psv-)potential, and never passive (see also §2.1.2).

The differences between low and high potential readings of *-rare* also show up in their translations and paraphrases in English. Low psv-potentials, like those in (5), are (roughly) equivalent to English V-*able* (e.g, *wear-able, read-able, reliable*) and contain a 'root' modality MOD$_{ability}$. High psv-potentials, as in (6), contain an 'alethic' modality MOD$_{possibility}$, and are **not** translatable with *V-able* in English. The modal part of the high potential is covered by the alethic use of *can* in English (Cinque 1999) or by the bi-clausal (non-passive) '*it is possible to ...*' as in '*it is possible to read books in that shop.*' Our empirical investigation will lead to the conclusion that the high modality in Japanese is also basically bi-clausal, embedding a much larger structure than the low modality, a (non-finite) proposition (roughly) of the same size as the *to*-infinitival complement of *possible* in English.

1.2 Theoretical assumptions

The assumptions that guide our investigation and analysis broadly fall within the Merge-based Minimalist tradition that includes strong decomposition, the search for (potentially universal) hierarchies (i.e. Cartography),[9] Antisymmetry (linear order reflects hierarchical order, in accordance with the LCA, with Spec H Complement as universal order),[10] and Morphology-as-Syntax,[11] with (remnant) phrasal movement the norm, and head movement severely restricted, without any of the postsyntactic adjustments of Distributed Morphology.[12] Furthermore, the model hypothesizes that the interfaces of syntax with phonology and the interpretive component are (as) direct (as possible). This forces one to constantly reconsider and adjust the syntactic structures we have grown familiar with.

The atoms that enter into Merge are (single) features (Kayne 2005:212, Koopman 2003), which have lexical properties, specifying denotation, category (label), phonological properties ranging from overt segmental and autosegmental units to **silence** (i.e. absence of any phonological realization), and selectional properties which specify their local syntactic environments−elements immediate to their right (i.e. complements) and to their left (i.e. Spec).

Selectional properties drive the syntactic derivation. We assume a principle of Locality of Selection,[13]which states that selectional properties must be locally 'satisfied,' i.e. be in a Merge configuration with the selector. This implies a selectee can only be a complement or a Spec of a selector. (Here, we depart from the current literature in not adopting the looser notion of Agree.) Suffixes merge at specific height (depending on the size of their complements), and select for a phrase to their left, forcing movement to the left of the suffix: this can be implemented either as an (structure building) epp-feature of the head requiring a specifier and driving I-merge (phrasal movement) or as arising from the properties of a head immediately

[8]Here and below, we give both a literal translation with *can+passive* for the high psv-potential, so as to show the source of the element corresponding to the grammatical subject position (marked with *-ga*), and an approximate translation with the bi-clausal (non-passive) '*possible to...*.'

[9]See Rizzi (1997, 2004a,b,c), Cinque (1999, 2005, 2006, 2009, 2023), and the Cartography of Syntactic Structures series.

[10]Kayne (1994, 2000, 2005, 2010, 2019).

[11]Koopman (2005, 2017), Collins & Kayne (2020), Caha (2018, 2009, 2020).

[12]Halle & Marantz (1993), Embick (2007), Embick & Marantz (2007), Embick (2010), Embick & Noyer (2007), Bobaljik (2017).

[13]Following Sportiche (2005), Sportiche, Koopman & Stabler (2013).

above the suffix (Kayne 2017b, Koopman & Szabolcsi 2000).

Structures are built bottom-up derivationally, via E(xternal)-merge (base generation) and I(nternal)-merge (movement), with cyclic spell-out and interpretation. The Extension condition (Chomsky 2000) restricts Merge to the root. I-merge is constrained by Relativized Minimality (cf. Rizzi 1990, 2002), Locality (Attract closest). Anti-locality (too close movement is blocked; Abels 2003, Kayne 2005) further regulates movement. There are privileged positions where silent DPs (heading a chain) can occur (PRO in Spec, TP headed by specific C). Finally, we assume very standard accounts for scope and binding, which interact with the structure in the usual way. Given these assumptions, properties of potential and passive constructions should fall out in a modular fashion from the interaction of the lexical properties of *-rare*, its local environments, and the general principles of structure-building and constraints.

Since we adopt Antisymmetry, there is a question of how we represent the surface 'head finality' of Japanese. We depart from the standard way of representing Japanese as being underlyingly head final. Head final surface structures are the output of the syntactic derivation, with verbal suffixes, which are clearly phrasal affixes, merged in the spine, and requiring a phrasal verbal constituent to their left. This movement, as we will see, feeds movement to the nominative.[14] We will represent object(s) in the big VP as preceding the (big) V and treat Case markers and P/K (postpositions) as merged in the spine, attracting DPs to their Spec. Our assumptions about multiple nominative subject *ga*-markings are detailed in §2.2.

1.3 Organization

The paper is divided into twelve sections. §2 introduces the core empirical data. It introduces various oblique passives (i.e. goal, directional, and source, on the one hand, and instrumentals and locative adjuncts[15]on the other), and documents their respective passivisability in the passive construction (§2.1) and in psv-potentials (§2.2). This leads to the research questions about oblique passives in §2.3: why are some oblique passives possible in any psv-potential and passive construction, and others (locative adjuncts and instrumentals) are only possible in high *possibility* psv-potential constructions.

§3-§6 set the stage for an evaluation of our hypothesis that *-rare* in passives and psv-potentials is the same element. §3 analyzes the decomposition and forms of *-rare* in the two constructions. We show potentials are composed of MOD, which comes in two flavors, and passive *-rare*, which must be structurally adjacent to ensure the correct spell out. §4 establishes a single hierarchy for the various elements in question. Specifically, §4.1 establishes the location where core passive Voice is merged. §4.2 discusses how passive voice interacts with low and high *ga*-marked obliques, and §4.3 brings $\text{MOD}_{ability}$ and $\text{MOD}_{possibility}$ into the hierarchy. This leads to the overall syntactic hierarchy for the passive and psv-potential constructions (see (7)), with *-rare* mergeable in more than one position depending on the height at which the two potential modalities can merge, and where the different obliques merge:

[14]See Kayne (1994, 2003), Koopman & Szabolcsi (2000), Julien (2002), Koopman (2005), and Kim (2023) for Korean.

[15]The distinction between directional and locative adjunct PPs (event modifiers, 'state PPs') is important, as directional and locative adjunct PPs occur in different regions of the clause. Directional/"Path" PPs, establish a path relation between a theme and its (end or starting) location, while locative adjuncts locate the entire event including the external argument. These differences (often) correspond to different distributions, as can be clearly seen in Mandarin, where path PPs are in the postverbal domain (like resultatives), but locative adjunct PPs must precede the little *v*P.

(7) ... $\boxed{\text{MOD}_{possibility} > {}^{H}\text{Pass}_{rare}}$ ... Loc/Instr ... $\boxed{\text{MOD}_{ability} > \boxed{{}^{L}\text{Pass}_{rare}}}$ [$_{vP}$... Goal/Dir/Source...]

§5, discusses how (7) accounts for the research questions given in §2.3. §6 expands the hierarchy (7) by incorporating the perfect/progressive marker *-tei*. This provides further support for the modal hierarchy in Japanese, with perfect/progressive *-tei* merging above MOD$_{ability}$ and passive *-rare*, but below MOD$_{possibility}$ (i.e, MOD$_{possibility}$.. > Perf/progr$_{tei}$.. > MOD$_{ability}$>Pass$_{rare}$). This overall cartography yields the relative order of (E-)merge steps for the different elements involved in passives and psv-potentials. This constrains the analytical options for a unified analysis of *-rare*, which we motivate in §7.

§7.1 introduces the lexical properties of passive voice suffix *-rare* (Ishizuka 2010, 2012), and spells out the basic analysis that includes (i) what complements *-rare* can take (*v*P and HabP); (ii) properties about its Spec: *-rare* is a phrasal suffix and selects for a phrase containing a verb as its Spec. This forces IM of a phrase containing V; (iii) Anti-locality freezes the immediate complement of *-rare*, and makes the complement of the immediate head the closest verbal phrase, movable to Spec, *rare*. We thus show that passive voice *-rare* always has the same syntactic effects. It does not directly interact with argument structure, but rather serves as a vehicle that, by attracting a verbal phrase to its Spec, carries arguments (or adjuncts) up in the derivation, feeding into further I-merge (promotion). Our analysis is thus basically a 'smuggling' type analysis (Collins 2005b, 2024, Belletti & Collins 2021). In §7.2, we run through some derivations, and show how we derive the *ga*-marked theme and oblique (i.e. goal, directional, and source) in passives, low 'ability' psv-potentials, and locative and instrumental nominatives in high 'possibility' psv-potentials.

In §8, we provide independent support for the proposed hierarchy and address further details of the derivations. §8.1 establishes the hierarchy among high obliques as Loc>Instr>Manner based on neutral word order and Minimality Effects. In §8.2, we show that the implicit argument in the high psv-potential is in a non-thematic subject−*i.e.*A−position from which it can bind into locative and instrumental PPs and control. We also address further issues related to Binding Theory, and suggest our structures provide some new insights. Lastly, §8.3 shows that our proposed derivation yields an account for the scope pattern of nominative-oblique alternation of locatives, consistent with the *ga*-marked nominative being an A position.

§9 extends the proposed analysis for a unique *-rare* to active potentials. We show how a simple switch in the local merge order of Passive$_{rare}$>MOD straightforwardly derives active potentials, and yields an account for the form and presence of a passive morpheme in active potentials.

§10 discusses if and how the unified analysis can be extended to other *-rare* constructions in Japanese−subject honorific (§10.1) and middle constructions (§10.2). §10.3 discusses how our account can be extended to capture the A properties of tough-movement.

§11 takes stock and discusses future directions. In §11.1, we turn to comparative syntax−ciCewa (Bantu) in §11.1.2 and Malagasy (Austronesian) in §11.1.1, which bring further support for our proposal that in Japanese locatives and instrumentals are merged *above* active voice and passive voice (contra Pylkkänen 2008). There must, however, also be a subject (and object) position *above* locative and instrumental PPs, just as Pylkkänen (2008) proposes. In §11.2, we show how this falls out from Kayne's proposal for the way PPs are built. Finally, in §11.3, we construct a tentative typology of oblique passives based on the languages discussed in the paper. §12 concludes.

2 Promotion of Obliques in Passives and Psv-Potentials

2.1 Promotion of obliques in the passive

This section introduces two kinds of obliques in Japanese and their distribution in the passive: 'low' obliques (goal(IO), directional, source) vs. 'high' obliques (locative adjunct, instrumental, manner). For high obliques, we will mainly focus on locatives and instrumentals, since they allow us to compare Japanese with instrumental and locative passives in Malagasy (§11.1.1) and ciCewa (§11.1.2). To a lesser extent, we also discuss manner obliques.

2.1.1 Goal, directional, and source passives

Ishizuka (2010, 2012) argues that Japanese, like English, has oblique passives (or pseudopassives), as in (8).

(8) **Goal, directional & source passives**

a. Sono ko -ga (Ken-ni) ame -o watas -are -ta.
that child -NOM (Ken-DAT) candy -ACC hand -(R)ARE -PAST
Lit. 'That child was handed candies to (by Ken).' Goal (*ni*)

b. Uti-no kuruma -ga (kodomo-ni) boru -o butuke -rare -ta.
we-GEN car -NOM (child-DAT) ball -ACC throw.at -RARE -PAST
Lit. 'Our car was thrown a ball at (by a child).' Directional (*ni*)

c. Kono keimusyo -ga (syuzin-ni) nige -rare -ta.[16]
this prison -NOM (prisoner-DAT) escape -RARE -PAST
Lit. 'This prison was escaped from (by prisoners).' Source (*kara*)

Oblique passives are difficult to diagnose because, contrary to English, the postposition is not stranded but absent. This is in fact a very general property of Japanese:

(9) "Oblique" DPs alternate with full PPs only if they combine with *-ga* or are relativized.[17]

We take the oblique DPs in Japanese (a subset of those marked with Ps like *-ni*, *-de -kara* elsewhere) to be Caseless DPs, i.e, smaller than full PPs. This allows analyzing movement of the oblique DP to nominative *-ga* as Case-driven A-movement. An additional P layer in full PPs is responsible for Case of the oblique DP, in which case there is no interaction with any A-movement. Throughout this paper, we notate Caseless oblique DPs lacking the P layer as DP_{goal}, DP_{source}, DP_{loc}, DP_{instr} and so on (see also fn 41).

It turns out that not all types of obliques can be passivized: only Goal(IO), Directional, which take P *-ni* 'to, at,' and Source, which takes P *-kara* 'from' as full PPs, can.[18] We call these 'low' obliques because they are merged below *v*, within the extended domain of the big VP. In fact, these obliques are low enough in the

[16] Without the overt P, the exact source of the nominative subject in (8-c) is difficult to identify. Since the preferred interpretation of the subject in (8-c) is a source, we assume that the nominative subject is underlyingly an oblique source.

[17] This effect also holds for possessor-raising constructions in Japanese, and more broadly crosslinguistically: genitive Case/linkers appear to be absent when the possessor raises to an argument position (Kuno 1973, Kuroda 2012, Landau 1999, and many others).

[18] Possessor-raising is possible in regular passives (see Kubo 1992, Ishizuka 2010, 2012) and (inanimate) inalienable (part-whole) possessors can be promoted to the nominative subject in psv-potentials (see also Nakamura & Fujita 1998).

hierarchical structure to be c-commanded by the passive voice and to undergo passivization (i.e, $\boxed{^{L}\text{Pass}_{rare}}$ [$_{vP}$... Goal/Dir/Source...]; see also (7)).

2.1.2 No locative, instrumental, manner passives

There is a second class of oblique nominatives (marked with the K/P *-de* elsewhere), consisting of locative (adjuncts), instrumentals and (some) manners. As is well known, instrumentals and manners require an external argument, and locative adjuncts are higher than instrumentals and manners (see §8.1 for their relative order). We refer to these as 'High' obliques. In passives, high obliques cannot map onto the nominative:

(10) **No locative, instrumental, manner passives**

a. *Kono kafe-ga (oisii keeki-o) tabe -rare -ta.
this cafe-NOM (tasty cake-ACC) eat -RARE -PAST
Int. '*This cafe was eaten (tasty cake) in.' $\boxed{\text{Location } de}$

b. *Kono pen-ga (tegami-o) kak -are -ta.
this pen-NOM (letter-ACC) write -(R)ARE -PAST
Int. '*This pen was written (a letter) with.' $\boxed{\text{Instrument } de}$

c. *Kono hooho-ga (taizyu-o) hayaku heras -are -ta.
this method-NOM (weight-ACC) quickly reduce -(R)ARE -PAST
Int. '*This method was reduced (weight) quickly with.' $\boxed{\text{Manner } de}$

However, these high obliques can be promoted to the nominative subject in high *possibility* psv-potentials, as we will show in §2.2.2 (see (15)). Our discovery that these obliques behave differently in passive and potential constructions plays a central role in our paper.

2.2 Ga-marking and Promotion of obliques in the psv-potential

Before we can explore the behavior of low and high obliques in low and high psv-potentials, we clarify the situation around subject *ga*-marking in Japanese. As is well known, Japanese allows multiple nominative subjects, and the literature recognizes two types of nominative *-ga*: exhaustive-listing *-ga* and descriptive *-ga* (eg. Kuno 1973, Kuroda 1965, 1992, p.49). Kuroda (1992, p.279) notes that exhaustive-listing *-ga* is a root phenomenon, and shows that when exhaustive *-ga* is embedded, for example under *mosi* 'if,' *-ga* no longer gets an exhaustive interpretation. This shows that nominative *-ga* and exhaustivity are not inherent properties of *-ga*, but must co-occur in root environments (for reasons that need to be elucidated). Exhaustivity is said to only be possible for the leftmost *ga*-marked DP in a sequence of *ga*-marked DPs. Kuroda (1992) analyzed multiple nominatives as arising from cyclic *ga*-assignment within a theory of Japanese Case that is perhaps closest to the case in tiers proposal of Yip, Maling & Jackendoff (1987). We incorporate the generalizations for subject *-ga* that are relevant for our paper in (11). We assume that *-ga* can enter the derivation at 2 levels, quite early, as it is compatible with the position of indefinite subjects in the hierarchy of high indefinite obliques (see (53) §8.1). *-ga* can also enter the derivation quite high, in Spec, TP, just below the C region.

(i) Kono ringo-ga kawa-o tabe-rare-ru.
this apple-NOM skin-ACC eat-RARE-PRS
✓'This apple's skin is edible.' $\boxed{\text{Inalienable, part whole}}$

(ii) *Ken-ga ringo-o tabe-rare-ru.
Ken-NOM apple-ACC eat-RARE-PRS
Int. '*Ken's apple is edible.' $\boxed{\text{Alienable, possessor}}$

In this paper, however, we focus on (non-genitive) obliques/PPs, and leave possessor raising out of the discussion.

It must combine with the exhaustivity operator in root clauses, but not in embedded clauses. This results in having multiple *-ga*'s on different noun phrases, though on a single DP only *-ga* can appear (i.e, **DP-ga-ga*).

(11) (i) [$_{root}$Exhaustive ga$_{nom}$ ...... ga$_{nom}$ ...] (ii) [$_C$ ga$_{nom}$ ..... (Indefinite) ga$_{nom}$ ...]

The root/non-root distinction helps determine if movement to exhaustive *-ga* is A-movement (as in cf. Saito 1982) or A'-movement (as in Inoue 2004:98). Since the (high) nominative case *-ga* in non-root clauses is not exhaustive, we assume the high *-ga* does not inherently bundle exhaustive focus. We thus analyze all movement to *-ga*, including to the high nominative *-ga* as involving a step of (normal Case driven) A-movement. This is supported by additional arguments in §8.2 and §8.3.

2.2.1 Goal, directional, and source passives in Low psv-potentials

Given MOD>Voice$_{passive}$ and bottom-up structure building, we expect not only themes (see (5)) but also low obliques to map onto the nominative in low *ability* psv-potentials, just as in passives, like (8).[19] Indeed this is the case, though oblique nominatives are not always easy to construct. It turns out that in root sentences low oblique psv-potentials are restricted to exhaustive *-ga*.[20] An additional difficulties for constructing low psv-potential sentences come from competing parses. With human nominative subjects, like (12-a), there is a very strong preference to interpret sentences as *active* potentials, where the nominative subject is interpreted as an agent or experiencer. A further difficulty that may play a role are the many functions of Japanese *-ni*, as a dative subject, a goal, a directional, or a *by*-phrase. (12) illustrates oblique passives in root clauses with exhaustive *-ga* and accusative (not nominative) objects.[21]

(12) Low psv-potentials with low oblique nominatives

a. [Context] *Ken is supposed to give candies to children in the room, but all of them are older and look scary to him, except one child.*
Sono ko -ga (?Ken-ni) ame -o watas -e -ru.
that child -NOM (Ken-DAT) candy -ACC hand -(RAR)E -PRS
≈ '(Only) thát child can be handed candies to (?by Ken).' Exhaustive;Goal (*ni*)

b. Kono kabe-ga (?Ken-ni) boru-o butuke -rare-ru.
this wall-NOM (Ken-DAT) ball-ACC throw.at -RARE-PRS
≈'(Only) this wall can be thrown a ball at (?by Ken).' Exhaustive; Directional (*ni*)[22]

c. Kono mado-(kara)-ga (?sono ko-ni) nige -rare-ru.
this window-(from)-NOM (that child-DAT) escape -RARE-PRS

[19]English allows for a few similar cases where the P is absent, e.g, *reliable vs rely on* (see Kayne 1984b, p.140-142, for a list and references). Note that (at least some) English native speakers tolerate pseudopassives in V-able constructions: *?This prison is escapable from, ?John is talkable to*, which for them contrasts with **this cafe is drinkable in*. We are not aware of systematic work on intra- and inter-speaker variation in this area.

[20]This does not hold for regular oblique passives, nor for theme psv-potentials

[21]We restrict our examples to accusative-marked objects to avoid complications with the Nom-Acc Case alternation on objects. Much of the Japanese literature on potentials has focused on nominative vs accusative objects, in particular with respect to their different scopal properties. See §8.3 for references and similar scope interactions between oblique nominatives and their corresponding full PPs in potentials.

[22]A permission reading (i.e. close to "legally" possible) is readily accessible for (12-a) & (12-b). Under this reading, a *ni* (by)-phrase is *excluded*. This also holds for (13-a) and (13-b). Interactions with Perfective/Progressive *-tei* show that permissive *-rare* involves the high *possibility* potential with (*rare*>*tei*) (see §6 and Appendix B for more discussion).

≈'(Only) this window can be escaped from (?by that child).' | Exhaustive; Source (*kara*)

In embedded contexts, nominative obliques do not need to get an exhaustive reading, confirming Kuroda (1992)'s generalization about exhaustive *-ga* being restricted to root clauses.

(13) (Embedded) Low psv-potentials with low oblique nominatives (non exhaustive)

a. [Mosi kono ko -ga (?Ken-ni) ame-o watas -e -ru -nara] . . .
if this child -NOM (Ken-DAT) candy-ACC hand -(RAR)E -PRS -then
≈'If this child can be handed candies to (?by Ken), . . . ' | not exhaustive: Goal (*ni*)

b. [Mosi kono kabe-ga (?Ken-ni) boru-o butuke -rare-ru -nara] ...
if this wall-NOM (Ken-DAT) ball-ACC throw.at -RARE-PRS -then ...
≈'If this wall can be thrown a ball at (?by Ken), ... '
≈ 'If it is possible (for Ken) to throw a ball at this wall, ...' | not exhaustive; Directional (*ni*)

c. [Mosi kono mado-(kara)-ga (?sono ko-ni) nige -rare-ru -nara]...
[if this window-(from)-NOM (that child-DAT) escape -RARE-PRS -then]...
≈'If this window can be escaped from (?by that child), ...' | not exhaustive; Source (*kara*)

It is not the case, however, that *-ga* in root potentials is always exhaustive-listing, as theme marked with *-ga* in (5) can be taken as descriptive *-ga* (especially without demonstratives) in root environments. Thus oblique psv-potentials in roots must get an exhaustive reading, while theme passives do not need to.[23]

(12-c) and (13-c) finally show that *-kara* 'from' behaves differently from *-ni* in allowing either full PP pied-piping or DP-*ga* in low psv-potentials.[24]

2.2.2 High psv-potentials

As we pointed out, low psv-potentials with theme subjects can often be reasonably translated into English *V-able*, but this is not the case for **high psv-potentials**, which paraphrase closer to '*it is possible to*,' or with English (possibility) *can*. The interpretive differences between the low and high psv-potentials suggest that there are at least two modal flavors available, $\text{MOD}_{possibility}$ and $\text{MOD}_{ability}$. According to Cinque (1999:76), these are merged at different heights in the structural hierarchy (the 'possibility' modal is (much) higher than the 'ability' modal, ($\text{MOD}_{possibility}$ ..>.. $\text{MOD}_{ability}$). We will show below this holds for Japanese as well. We call the psv-potentials containing the high modal−'possible'−high. We will show in §4.3 that the two modalities in fact can cooccur, further supporting the presence of the two modality heads in Japanese.

2.2.3 Goal, directional, source (i.e, low oblique) passives in High psv-potentials

Similar to low psv-potentials, theme and low obliques like goals, directionals, and sources, can be promoted to the nominative subject in high psv-potentials, but a dative *ni* by-phrase is (generally) not allowed. The external argument of the main predicate is, therefore, obligatorily silent. Furthermore, it must receive a

[23]Interestingly, Kuroda (1992:ch.8) reports that the same generalization holds for tough movement constructions. We do not have an account for why only exhaustive/high *-ga* is available for oblique psv-potentials in root environments while theme psv-potentials can use low *-ga* in root and non-root environments. This could follow if oblique K can be shown to enter the derivation *above* the low *-ga*, leaving low *-ga* too deeply embedded for Caseless obliques. We leave this for future investigation.

[24]We note differences in judgments here. The first author considers sentences with overt *kara* like (12-c) acceptable in psv-potentials and in tough movement, but not in passives. Kuroda (1992, ch.8) rejects a psv-potential sentence with an overt P *kara* (DP-*kara-ga*), but accepts them in adjectival tough-constructions. We will not address this issue any further here.

generic human interpretation like 'one, you, people, we,' as shown in (14) (see Kanno 1992, Fukuda 2025, among others).

(14) Sono supaisu-ga (*Mei-ni) kono mise-de kaw -e -ru. $\boxed{\text{Theme } (o)}$
that spice-NOM (*Mei-DAT) this shop-LOC buy -(RAR)E -PRS
'That spice can be bought at this store (*by Mei).'
≈ 'It is possible (*for Mei) to buy that spice in this store.' 'One can buy that spice in this store.'

The differences between high and low psv-potentials are not restricted to a difference in modal flavor and the (in)availability of a dative *by*-phrase. Importantly, a wider variety of obliques can map onto the nominative subject in high psv-potentials than in regular passives, as we will show in the next section.

2.2.4 Locative, instrumental, manner (i.e, high oblique) passives in High psv-potentials

Promotion to the nominative subject is not only possible for themes and low obliques, but also for high obliques, such as locative adjuncts (15-a), instrumentals (15-b), and manners (15-c), which must get an exhaustive reading in root clauses, but do not have to get such a reading in embedded complements.[25] As (10) shows, this is excluded in passives, thus distinguishing high psv-potentials from regular passives.

(15) **High psv-potentials for high obliques (locative, instrumental, manner)**

a. Kono kafe-ga oisii keeki-o tabe-rare-ru.[26] $\boxed{\text{Location } (de)}$
this cafe-NOM tasty cake-ACC eat-RARE-PRS
Lit. 'This cafe can be eaten tasty cake (in).' ≈'It is possible to eat tasty cake in this cafe.'

b. Kono pen-ga akai zi -o kak -e -ru.
this pen-NOM red character -ACC write -(RAR)E -PRS
Lit. 'This pen can be written red characters (with).'
≈'It is possible to write red characters with this pen.' $\boxed{\text{Instrument } (de)}$

c. Kono hooho-ga taizyu-o hayaku her.as-e -ru.
This method-NOM weight-ACC quickly reduce-(RAR)E -PRS
Lit. 'This method can be reduced weight quickly (with).'
≈'It is possible to reduce weight quickly with this method.' $\boxed{\text{Manner } (de)}$

In sum, the following properties characterize **high psv-potentials**:

[25] Temporals can also be marked nominative, and require an exhaustive focus reading in root psv-potentials, as in (i).

(i) *Context: Once a month cake is served in the library.*

Ashita -ga tosyokan -de sono keeki -o tabe -rare -ru.
tomorrow -NOM library -LOC that cake -ACC eat -RARE -PRS
'It is possible to eat that cake tomorrow (tomorrow's (monthly) cake) in the library.'

We set temporal nominatives aside, and concentrate on locative and instrumental passives which, contrary to temporal nominatives, are known to occur in other languages (See §11.1.1).

[26] There are two arguments against a possessor raising analysis of (15-a), with *this cafe* raising out of the DP: *[[kono kafe-NO] oisii koohi]* 'this cafe's tasty coffee.' First, because the interpretation of (15-a) is also felicitous in context where the tasty cake is made not in 'this cafe' but somewhere else, and then brought to 'this cafe.' Secondly, because a locative passive is also available in sentences like (i) without an overt direct object.

(i) Kono tosyokan-ga (yoku) nemur-e -ru.
this library-NOM (well) sleep-(RAR)E -PRS
Lit. 'This library can sleep (well) (in).'≈ 'It is possible to sleep (well) in this library.'

In (i), there is no possible source for possessor-raising, which guarantees that this is a case of a locative psv-potential.

(16) (i) The high modality $\text{MOD}_{possibility}$ and passive *-rare* are part of the structure.

(ii) Locative, instrumental, manner passives (high oblique passives) are possible (as well as theme and low oblique passives).

(iii) The oblique Case/postpositions *de*, *ni* must be absent, as in passives, while *kara*, as in DP_{source} *kara-ga*, is possible in potentials, unlike in passives.

(iv) The external argument of the lexical V is obligatorily silent.

(v) The implicit argument must get a [+human] generic interpretation ('people, one, we, you'). This is reminiscent of impersonal *si* in Italian (Cinque 1988), the impersonal passive in Turkish (Legate et al. 2020) and the accusative passive in Maasai (Greenberg 1959).

The table below summarizes the distribution of nominative DP-*ga* in psv-potentials and passives.

Table 1: **Distribution of the nominative subject**

Derived from	Alternates with K/P	High psv-Potential	Low psv-Potential	Passive
Theme	*o*	ok	ok	ok
Goal	*ni*	ok	ok	ok
Directional	*ni*	ok	ok	ok
Source	*kara*	ok	ok	ok
Location	*de*	ok	*	*
Instrument	*de*	ok	*	*
Manner	*de*	ok	*	*

2.3 Research questions about oblique passives

We can now formulate the core research questions about oblique passives:

(17) a. Why are **Goal and Directional** passives **possible** in both Passives & Psv-Potentials?

b. Why are **Locative and Instrumental** passives **excluded** in Passives
but **possible** in *High* Psv-Potentials?

High obliques are above low obliques, and the question is how this height difference interacts with the passive and the psv-potential. We assume that there are at least two locations for passive Voice in the clausal spine: one contained in the regular passive ($^{L}\text{Pass}_{rare}$), and the other introduced higher and dependent on the high modality head, $\text{MOD}_{possibility}$ ($^{H}\text{Pass}_{rare}$).[27]

Our proposal is that high obliques like locatives are merged **above** $^{L}\text{Pass}_{rare}$, but **below** $^{H}\text{Pass}_{rare}$.

(18) .. $\boxed{\text{MOD}_{possibility} > {}^{H}\text{Pass}_{rare}}$ .. Loc/Instr > ... $\boxed{\text{MOD}_{ability} > \boxed{{}^{L}\text{Pass}_{rare}}}$ [$_{vP}$ Goal/Dir/Source..]

This allows answering the research questions (17-a) and (17-b), since passivization requires c-command by passive Voice. We next move towards motivating this hierarchy in §3 and §4, and provide the answers in §5.

[27]The $\text{MOD}_{ability}$ dependent $^{L}\text{Passive}_{rare}$ might in fact be slightly higher than the canonical passive Voice (see fn 34 and (93) for possible evidence), in which case there would be (at least) three heights for *-rare*. However, since this finer hierarchy does not directly affect our argumentation, we have omitted it, pending further investigation.

3 Structural Make-up of Potentials and Their Forms

In this section, we make our assumptions about the passive and potential forms explicit. This will enable us to first determine the cartography of psv-potentials and passives with respect to the thematic hierarchy, and low and high obliques (§4.2). This will then lead us to the answers to the questions given in (17).

3.1 The forms of the suffix *-rare* in passives and psv-potentials

Given the passive properties observed in psv-potentials, we concluded in §1.1 that *-rare* in psv-potentials is the passive voice suffix *-(r)are*. In addition, a silent MOD (either MOD$_{ability}$ or MOD$_{possibility}$) is part of the structural make up of psv-potentials.

We further saw in §1.1 that the form of *-rare* in (psv-)potential constructions depends on the final segment of the verb stem in contemporary standard Japanese (see (4) and (5)). If the verb ends in a consonant, a short form *-e* occurs (*ar* is absent). If it ends in a vowel, the full form *-rare* occurs.

Final segment of V	Passive	Potential
[$_{Verb}$... V] #	rare	rare
[$_{Verb}$... C] #	are	$\boxed{\text{(ar)e}}$

We follow the standard account by Itō & Mester (2003) and many others, according to which the choice of the short potential form is dependent on phonological properties of the stem.[28]

Since the short *-e* form is only available in the potential (but not in the passive),[29] the trigger of its selection must be linked to the presence of the silent MOD. The phonological sensitivity shows that the stem and Pass$_{rare}$ & MOD−**must be structurally 'close.'** In keeping with the one-feature-per-head decomposition hypothesis, we assume MOD and Pass$_{rare}$ each project in the syntax, and have to be in a local head complement relation as a precondition for the selection of the right allomorph.[30] Since MOD is silent in Japanese, both orders MOD>Pass and Pass>MOD are in principle compatible with this condition. We have seen so far in §1.1 that the order of MOD>Pass results in low psv-potentials. In §9, we will show that the other possible order of Pass>MOD, with modalization preceding passivization, is also in effect, accounting for the presence of the passive suffix in the active potential.

3.2 Crosslinguistically, psv-potentials consist of two heads

When comparing Japanese *ki-rare* 'wear-pass' with the English equivalent *wear-able*, we observe the Japanese form spells out passive Voice, but not the modality, whereas the English form spells out the modality, but

[28]In the contemporary colloquial register, potential *-rare* can optionally be realized as *-r*(ar)*e* after V-final verb stems (aka. *ra-nuki kotoba* 'lit. *ra*-taken out words'). Itō & Mester (2003) present an OT analysis of the optional variation by means of paradigm contrasts. Sano (2015, fn.6) presents the following list of the (somewhat perplexing) variables identified in the literature governing the variation between *re* and *rare*: (i)-*(r)e* is (i) more compatible with affirmative contexts than with negative contexts; (ii) occurs only with short verb stems but not with compound, auxiliary, or causative verbs; (iii) more frequent in main clauses; (iv) more compatible with verb stems that end in [i] rather than in [e]; (v) is preferred by younger speakers and female speakers. Dealing with this phenomenon is outside the scope of this paper: further research is necessary in light of the new generalizations we establish.

[29]The first author agrees with Shibatani (1985) that the exponent of the middle/spontaneous suffix can be *-e* as well as *-(r)are* (*hag-are-ru* '(spontaneously) come off,' as exemplified in (69)). Kanno (1992) only allows *-e* for the middle/spontaneous suffix.

[30]We suggest Pass$_{rare}$ & MOD might 'compress' or 'bundle' at the end of the derivation to ensure the proper selection of the form based on the final segment of the verb stem. This implies that rather than starting out as a single head *-rare* with both a modal and a passive feature, and ordering their respective probing features to mimick the two merge orders, we end up with a feature complex created in the syntax from the merge orders representing the order of operations. We return to this in §D.

not passive Voice.[31] This suggests that a modal and a passive component are involved in both languages, with different parts spelled out or silent (but both heads are active: in the form of the modal interpretation in Japanese, or as reflected by passive properties in English). We take this to show that potential constructions are crosslinguistically built up out of the same parts—a modal head and a passive Voice, but languages may differ only as to which part is spelled out (and as to the derived morphosyntactic category; e.g, *-able* forms an A, but *-rare* forms a (stative) V).

This hypothesis leads to the expectation that a third pattern should occur, with some languages spelling out both the modal and the passive for potentials. This appears to be the case, for example, for Turkish where both the passive morpheme *-(I)l/n* and the modal *-(y)Abil* are used in potentials (e.g, *yaz.ıl.abil.ir* 'write.PASS.MOD.AOR 'writable'; Oltra-Massuet (2008:18) and Lipták & Kenesei (2017:69)).[32]

4 Hierarchies 1: A (Partial) Cartography

We now turn to the hierarchical landscape surrounding *-rare*: this will inform the relative order of Merge operations, provide an answer to our research questions, and further constrain the understanding of where passive Voice can be merged. We do so in several steps. We first discuss what the syntax can tell us about the distribution of passive voice, the relevant modality, low and high obliques, and thematic structure. In order to understand what merges where, i.e. what the order of merge operations is, we address the following questions. This is needed for testing if we can motivate a single *-rare* for the passive and (psv-)potential.

- Where in the clausal hierarchy can passive voice be merged, especially with respect to low and high obliques? (see §4.1, and §4.2.)
- Where are the modalities that make up the high and low psv-potential, i.e. *possibility* and *ability*, merged? (See §4.3 for arguments that they occur in distinct regions in the clause.)
- What do the merge positions of the modal heads entail for the merger of passive voice *-rare*?
- How do the findings above account for the possible or impossible promotion from low (goal, directional, and source) and high (locative and instrumental) obliques? (See §4.3)

4.1 Where is Passive Voice merged in the passive construction?

Canonical passives give an indication about the **minimal height** passive voice *-rare* can be merged, but not about the **maximal height**. As is well established, passive *-rare* in Japanese (minimally) merges with transitive verbs or a subset of unergatives (provided they have more than one shell, Ishizuka 2012:42). It cannot merge with pure unaccusatives (eg, **oti-rare* 'fall-rare,' **tuk-are* 'arrive-(r)are', which both take a directional *-ni* complement in the active), nor with unmodified unergatives.

(19) *Kawa-ga (Ken-ni) **oti-rare** -ta.
river-NOM (Ken-DAT) fall-RARE -PAST
*'The river was fallen into (by Ken).'

(20) *Umi-ga (Ken-ni) **tuk -are** -ta.
sea-NOM (Ken-DAT) arrive -(R)ARE -PAST
*'The sea was arrived at (by Ken).'

[31] See Koopman (2021b) for arguments that English Passive Voice is always silent.

[32] The passive suffix takes the form *-(I)l* after consonants, *-n* after vowels and *-(I)n* after laterals, where (I) stands for a vowel, which harmonizes to the closest vowel in the stem (Kornfilt 1997, Murphy 2014). The modal *-(y)Abil* consists of a harmonizing vowel A and the light verb *bil*, derived from 'to know.' Under negation, the verb *bil* is absent (Kornfilt 1997, 374).

We take this to show that passive voice selects for a projection that introduces an external argument as its complement (i.e. either little v_{cause} or v_{do}), only because if it were to merge directly with a big VP, it would be hard to understand why unaccusatives cannot combine with passive voice. This yields the following (quite simplified) structure, with "big VP" in fact standing for a much larger constituent.

(21) Minimal height where passive voice can be merged

Pass$_{rare}$ [$_{vP}$ ext.argument [$v_{cause/do}$ [$_{\text{“big”VP}}$ V .....]]]

Importantly, though this will only play a role from §7 on, we conclude that the external argument is indeed syntactically projected in the passive (see Angelopoulos, Collins & Terzi 2020, Collins 2005b, 2024, among others, for compelling evidence), but silent for the same reason as PRO is silent in some infinitivals.

4.2 Interaction of Passive Voice with low and high obliques

Next we consider if there is also a maximum height for merging Passive Voice in passive constructions. Important clues come from the kind of obliques that can or cannot be passivized. Recall that high oblique passives (eg, locative and instrumental) are only possible in psv-potentials but not in passives (see (17)).

We start by spelling out a necessary, but not sufficient, condition on passivizability (see also §1.1).

(22) Only elements c-commanded by Passive Voice can in principle be passivized.

This condition is not sufficient, as passivization is further subject to whatever property that allows oblique passives in a language, and to Attract closest, which restricts extraction to the closest DP. The latter depends on the derived syntactic structure of the big VP when passive Voice is first merged.

Since low obliques, such as goals and directionals (*ni*-marked as full PPs) and sources (*kara*-marked as full PPs), can be passivized in Japanese, they must be in the c-command domain of passive Voice. This gets independent support: goals, directionals, sources are below little *v* in Japanese (eg, Kishimoto 2008, Ishizuka 2019, Miyagawa & Tsujioka 2004), and more generally crosslinguistically.

(23) Goal, directional, source passives (simplified):

Pass$_{rare}$ [$_{vP}$ ext.argument [v [$_{\text{“big”VP}}$ ..Goal/Dir/Source...]]]

These obliques can all be promoted to the nominative, as can the goal in a double object construction, thus closely corresponding to pseudopassives in English.

In contrast, high obliques, such as locative adjuncts, instrumentals, and manners (all *de*-marked as full PPs), are not passivizable, regardless of the presence or absence of an accusative object. This would follow if they are not c-commanded by passive voice, as illustrated in (24).

(24) Excluded: locative, instrumental, manner passives.

... Loc/Instr/Manner ... >... **Pass**$_{rare}$ [$_{vP}$ ext.argument [v [$_{VP}$ .....]]]

This explanation requires that high obliques be merged *above* the little *v*P. This is a priori reasonable, given the fact that instrumentals and manners require an external argument, and locative adjuncts are merged higher than instrumentals (see fn 8.1 for further language internal support, and §11.1 for crosslinguistic support from Malagasy (Western Austronesian) and ciCewa (Bantu)). However, this assumption is incompatible with Pylkkänen (2008)'s proposal for high applicatives, which we address in §11.1.

4.3 The psv-potential: Merging the silent MOD and Passive Voice

We now turn to the potential construction and consider where the silent *ability* and *possibility* modalities are merged, and the implications for the merger of its accompanying passive Voice *-rare*. For this purpose, we turn to Cinque (1999:79), who identifies two distinct locations for these modalities in the functional hierarchy of the clause: one for root modalilities like 'ability,' and a much higher one for alethic modality like 'possibility.' This leads to the following hypothesis in Japanese psv-potentials as to where the different modalities could be merged: ... $\text{MOD}_{possibility}$ > ..... > $\text{MOD}_{ability}$...

These two modalities may cooccur:

(25) ?Sonnani nagai-aida Mei-o ais **-e** -tei **-rare** -ru (to-wa!)[33]
such long-term Mei-ACC love -(RAR)E -PERF/PROGR -RARE -PRS C-TOP
'(How surprising that) it is possible (for you) to have been able to love Mei for such a long time!'

Importantly, these modalities appear in two distinct regions of the clause, which can be separated by a number of functional projections, including the perfect/progressive marker *-tei* as in (25) (see also §6), which turns out to be relevant in accounting for the different behavior of high obliques (locatives and instrumentals) in regular passives and in psv-potentials.

As we have established in §3.1, psv-potentials span a silent modal MOD and passive voice. This yields the following merge locations for the MOD dependent passive Voice *-rare* in high and low psv-potentials:

(26) High and Low psv- potentials:

$$\ldots\ \boxed{\text{MOD}_{possibility} > {}^{High}\text{Pass}_{rare}} > \ldots\ldots\ldots\ldots > \boxed{\text{MOD}_{ability} > \boxed{{}^{Low}\text{Pass}_{rare}}} > vP\ \ldots$$

We take ${}^{Low}\text{Pass}_{rare}$—the complement of $\text{MOD}_{ability}$—to be the same passive voice as the one in the regular passive construction, that is, it takes *v*P introducing the external argument as its complement.[34] Significantly,

[33] Some speakers might find (25) slightly degraded because of a general aversion (in Japanese) to having multiple instances of *-rare* in the same verbal complex. Changing the lower potential to *deki*, a suppletive potential form (do+$\text{Voice}_{passive}$+MOD), which behaves in all respects like the potential *-rare*, yields a fully acceptable sentence (i) (see Appendix A).

(i) Go-zikan-mo syucyu **-deki** -tei **-rare** -ru (to-wa!)
five-hour-also concentration -do.can -PERF/PROGR -RARE -PRS C-TOP
Lit. '(How surprising that) it is possible (for you) to have been able to concentrate for 5 hours!'

[34] It is sufficient for our purposes to treat the $\text{MOD}_{ability}$ as merging with the Pass_{rare} (in the regular passive), but this is certainly a simplification. For example, in the case of long passives, the ability modal passive can be above *sokone* 'fail,' but the regular passive cannot, as illustrated below (see also Appendix C).

(i) a. Keiyakusyo-ga {✓kawas-are-sokone / *kawasi-sokone-rare} -ta.
contract-NOM {✓exchange-PASS-fail / *exchange-fail-PASS} -PST
Lit. 'The contract {✓failed to be exchanged / *was failed to exchange}.' ✓fail> **Pass**>V *****Pass** > fail>V**

b. Keiyakusyo-ga (kare-ni) {*kak-e -sokone / ?kaki-sokone -r(ar)e} (-?tei) -ru.
contract-NOM him-DAT {*write-MOD -fail / ?write-fail -MOD} (-?PERF/PROG) -PRS
Lit. 'The contract {*fails to be able to write/?can fail to be written} (by him).' *fail>**MOD**>V ?**MOD**>fail>V

This might show that the (ability modal) *-rare* can be merged higher than *sokone* 'fail,' which is higher than regular passive Voice *-rare* (i.e, $\text{MOD}_{ability}$>${}^{L}\text{Pass}_{rare}$ > fail >Pass_{rare}>V: see (93)), suggesting there would be three merge sites for *-rare*. This topic needs to be further explored, but does not affect our argumentation, though it will help with the analysis of finer differences between passive and low psv-potentials.

there is an additional merge position for the passive voice, ^{High}Pass$_{rare}$, in the high psv-potential, which is made available by the MOD$_{possibility}$, as illustrated in (26).

In (24), we have suggested that locatives and instrumentals cannot be passivized because they are merged higher than *v*P, in fact above ^{Low}Pass$_{rare}$. We now see that high obliques *can* (in principle) be passivized as long as they are merged in the c-command domain of the ^{High}Pass$_{rare}$ (thus occurring only in high psv-potentials), as illustrated below:

(27) High and Low psv-potentials with obliques:

....$\boxed{\text{MOD}_{possibility}}$ > ^{H}Pass$_{rare}$ > ......**Loc/Instr**......> $\boxed{\text{MOD}_{ability} > \boxed{^{L}\text{Pass}_{rare}}}$ .....Goal/Dir/Source

5 Answers to Research Questions on Obliques

Given the interaction between passive Voice and high and low obliques, we now have our answers to questions (17-a) and (17-b), repeated below as (28-a) and (28-b):

(28) a. Why are **Goal and Directional** passives **possible** in both Passives & Psv-Potentials?
A: *This is because low obliques are part of the built structure below little v.*

b. Why are **Locative and Instrumental** passives **excluded** in Passives but **possible** in *High* Psv-Potentials?
A: *Locative and instrumental DPs are merged above Passive Voice and little v, not c-commanded by* ^{L}Pass$_{rare}$*: hence they are excluded in Passives and low psv-potentials;*
The MOD$_{possibility}$ *dependent passive Voice is merged* **above** *locative and instrumental oblique DPs, therefore they are in principle passivizable in high psv-potentials.*

Further, given the incremental bottom-up nature of structure building and derivations, the portion of the clause built below Pass$_{rare}$ is much larger for high psv-potentials, including, for example, accusative marked DPs, double object constructions, PPs, and as we will show in §6, perfect/progressive *-tei* (eg, (41)). This makes the complement of the high psv-potential *-rare* roughly comparable to the infinitival complement of *possible* in English. In other words, the high psv-potential is a bi-clausal structure in Japanese and English.

Importantly, the previous section has established that not only the two modalities are merged at different heights in different regions of the clausal structure, but so is the passive Voice, which is part of the (psv-)potential in Japanese (see (27)). While this is a novel proposal for the cases we discuss here, variation in height of merger within a language for passive-like voices is currently the standard approach for the difference between adjectival vs. verbal passives, lexical vs. syntactic causatives (see §10.2 for references and discussion). If height of merger is indeed responsible for the typological variation in passive-like constructions, we should also find languages where passive morphology can merge above instrumental and locative obliques, and thus allow instrumental and/or locative passives (see §11.3 for further discussion).

6 Hierarchies 2 (continued): Passive Voice *-rare* and Perfect/Progressive *-tei*

This section further sharpens the hierarchy established in (27) by bringing in the linear orders of passive Voice and perfect/progressive aspectual suffix *-tei*. The perfective/progressive[35] phrasal affix *-tei* is one of the functional categories that can occur between the high and low modal fields (see (25)), as shown in (29).

(29) Perfect /progressive in the psv-potential:

.. $\boxed{\text{MOD}_{possibility} > {}^{H}\text{Pass}_{rare}}$ > .. **Perf/progr**$_{tei}$ ... > .. Loc/Instr.. > $\boxed{\text{MOD}_{ability} > \boxed{{}^{L}\text{Pass}_{rare}}}$ [$_{vP}$ ..]

The two suffixes, *-rare* and *-tei*, can occur in either order, *V-(rar)e-tei* or *V-tei-rare* and correspond to different interpretations. This is expected if the former order *V-(rar)e-tei* contains the (low) passive voice merged with MOD$_{ability}$, while the latter order *V-tei-rare* contains the high passive voice that is parasitic on the presence of the high MOD$_{possibility}$.

In other words, *V-rare-tei* should be compatible with a low psv-potential reading and a passive reading (if MOD is not merged with Pass$_{rare}$). This is indeed the case, as shown in (30-a). (Note that a passive reading is unavailable in (30-b) because only potential spell out as *-e* with C-final verbs.)

(30) V-*(rar)e-tei*: **low psv-potential** or **passive** readings

a. Sono eiga-ga (Ken-ni) nan.zikan-mo mi **-rare** -tei -ru. $\boxed{\text{Theme}}$
that movie-NOM (Ken-DAT) many.hours-also watch -RARE -PERF/PROGR -PRS
✓'That movie can have been being watched for many hours (by Ken).' [Low psv-potential]
✓ 'That movie has been being watched for many hours (by Ken).' [Passive]

b. Sono hon-ga (Ken-ni) nagai aida yom **-e** -tei -ru. $\boxed{\text{Theme}}$
that book-NOM (Ken-DAT) long period read -(RAR)E -PERF/PROGR -PRS
✓'That book can have been being read for a long time (by Ken).' [Low psv-potential]

With *V-tei-rare* order, on the other hand, the regular passive reading should not be available because high passive Voice depends on the projection of the MOD$_{possibility}$, which forces the high psv-potential reading. This prediction is also born out:

(31) V-***tei-rare***: **high psv-potential** reading but ***no* passive** reading

Kono tosyokan-ga (osoku-made) hon-o yom **-dei** **-rare** -ru. $\boxed{(de)\text{-Location}}$
this library-NOM (late-until) book-ACC read -PERF/PROGR -RARE -PRS
✓ Lit. 'This library can have been reading books in (until late).' $\boxed{\checkmark\text{High psv-potential}}$
≈ 'It is possible to continue reading books (until late) in this library.'
Not ∗ 'This library has been read books in (until late).' $\boxed{\text{Excluded:Passive}}$

Importantly, given that high obliques are merged higher than low MOD$_{ability}$ and ^{L}passive Voice, we predict that a high psv-potential with locative, instrumental, manner nominatives is not available with *V-rare-tei* order, since high obliques are outside the scope of low passivization. This prediction is also borne out:

[35] Progressive *-tei* is closer in meaning to 'ongoing' and is compatible with stative verbs, which the English progressive is not.

(32) *Kono tosyokan-ga (osoku-made) hon-o yom **-e** -tei -ru. $\boxed{(de)\text{-Location}}$
this library-NOM (late-until) book-ACC read -(RAR)E -PERF/PROGR -PRS
Int. * 'This library has been readable books in (until late).' $\boxed{\text{* High psv-potential}}$

This is an example of textbook mirror order principle, where the morphology corresponds to the syntactic hierarchy, with the inner suffix combining with the V first and the outer suffix combining next.

(33) a. $\boxed{\text{V-}\textit{rare-tei}}$: Low psv-potentials & Passives:
⇒ Passivization **precedes** merging Perfect/Progressive.
b. $\boxed{\text{V-}\textit{tei-rare}}$: High psv-potentials
⇒ Passivization **follows** merging Perfect/Progressive.

Given the discussion above, we conclude that not only high obliques like locatives, instrumentals, and manners (in that hierarchical order, see §8.1) are found in the region between the two modalities, but so is the perfect/progressive *-tei*. Moreover, we can assume that high obliques are merged below the perfect/progressive. Locative and instrumentals behave like *v*P/VP adjuncts in English, stranding the auxiliaries under VP preposing. Putting it all together, our bottom-up derivations have to yield at least the hierarchical structure given in (34).

(34) High and low (psv) potentials with Perfect/Progressive: (initial version)
Nom$_{ga}$ >.. $\boxed{\text{MOD}_{possibility} > {}^{H}\text{Pass}_{rare}}$ > .. Perfect/progr$_{tei}$> ..loc/instr/mnr...> $\boxed{\text{MOD}_{ability} > \boxed{{}^{L}\text{Pass}_{rare}}}$
[$_{v\text{P}}$ ext.argument *v* [$_{\text{VP}}$ ....goal/dir/source ...]]

(34) is important with respect to the theoretical question where exactly the external argument is merged; the question how to differentiate between thematic and non-thematic syntactic subject positions, which will inform pieces of the syntactic derivation; and the question what the label of the complement ${}^{H}\text{Pass}_{rare}$ is. These questions directly impact our understanding of the effects of passive voice, not as directly interacting with argument structure, but as a formal mechanism, like a gondola, which carries arguments up to positions from where they can disembark onto the syntactic positions where they surface (see §7).

7 Analysis of Passive Voice

Given the relevant pieces of the hierarchical structure, which tell us about the order of operations, we now turn to our hypothesis about a unified analysis for *-rare*, and test it on the syntactic derivations. Under our analysis, psv-potentials contain the same passive voice *-rare* as (regular) passives, and *-rare* in passives and low/high psv-potentials differs only in terms of where it is merged in the structure (i.e, what its complement is), as schematized in (26), but the structural effects are always the same.

We can now compare our account with what has, to a large extent, been variations of the standard accounts of passive voice since LGB (Chomsky 1981), which predates VP shells. How do the standard accounts deal with the problem caused by the silence of the external argument and lack of intervention effects? In most accounts, the external argument is not projected in the syntax and therefore silent, and not intervening for movement of the object. This has been achieved in various ways. For example, as an effect of passive morphology, where morphology was hypothesized to be able to affect argument structure. Or in more recent Voice-based accounts (Pylkkänen 2008, Legate 2010, 2021, Bruening 2013, and others), where the external argument can fail to be projected in the syntax, because passive Voice, as opposed to active Voice, lacks a specifier. In this case, the semantics is held responsable for its interpretative effects.

Collins (2005b, 2024) and Angelopoulos, Collins & Terzi (2020), on the other hand, argue for a different syntactic solution, which exploits VP decomposition, and is close to the traditional theory in that it is based on a syntactically transitive structure: the external argument is projected in the syntax, as expected, but is always allowed to be silent for a principled reason. According to Collins, passivization (in English) is achieved via the phrasal movement of the participial VP (formed below little *v*) and attracted to the Spec, passive Voice. The moved participial VP contains the theme, thus bringing it closest to the nominative, and the external argument is stranded below the passive Voice. Thus the participial VP-movement 'smuggles' the object past the external argument, explaining why it does not intervene, or map onto the grammatical subject position (see (37)). Our investigation brings strong support for a Collins type analysis, since it allows for a unified analysis for the different *rare*-s. It also shows that analyses that interact directly with the projection of argument structure cannot lead to a unified analysis.

7.1 Properties of Passive Voice *-rare*

Adapting Collins' analysis to Japanese, Ishizuka (2010, 2012) proposes that *-(r)are* in the passive construction never directly reduces or introduces an argument. We extend Ishizuka's (2010, 2012) unified smuggling analysis for passive voice *-rare* in passives to psv-potentials and assume the following lexical properties of passive voice *-rare* in both constructions.

Complement of *-rare*: The passive voice *-rare* in passives and low psv-potentials takes 'little' *v*P as its complement (see §4.1), with *v*−an elementary predicate, such as *do* or *cause*−introducing the external argument of the verb (see Hale & Keyser 2002). We discuss how to label the complement of *-rare* in high psv-potentials in §7.2.6.

External argument introduced by 'little' *v*: The external argument is projected in the syntax, as it is required by v_{do} or v_{cause}. It is silent, for the same reason as the DP in the subject position of some infinitival complements with a silent infinitival C (cf. Collins 2005:p.104, Kayne 2006).[36] We consequently write it as PRO to bring out the parallelism between *-rare* and the silent infinitival C (cf. also §7.2.6 and §9).[37]

Specifier of *-rare*: Passive voice *-(r)are* always requires a (phrasal) verbal specifier, regardless of the height at which it is merged. This property can be implemented by an epp[+V] feature, which forces movement of a phrase containing a verb, and also derives the suffixal nature of *-rare* (see cf. Koopman & Szabolcsi 2000, Julien 2002, Koopman 2005, 2014, Collins & Kayne 2020, Kayne 2017b, for suffixes).

[36] The external argument can also be realized as a *ni*-phrase in passives and low psv-potentials (depending on the content of the DP), but *not* in high psv-potentials. Furthermore, the interpretation of PRO varies: existential in passives and low psv-potentials, but generic [+human] *one, you, people, we* in high psv-potentials. Though we are unable to enter into details for reasons of space here, we pursue the idea that (i) the *ni*-phrase is not in the same position as the silent external argument (see (58) for possible evidence), (ii) *ni* can only be merged in a restricted region above the (low) passive voice (reminiscent of Williams (2003) levels of representations), with the external argument I-merging with *ni*, and (iii) *ni* is not available for merge in the higher region of the clause. Thus, only derivations which involve the region where *ni* is available for merge will potentially allow for a *ni*-phrase. We leave the details for future research.

[37] We do not address the question if it should be PRO or pro (see Collins 2023:p.52ff for relevant discussion)

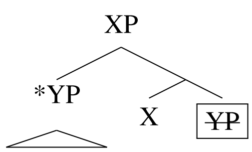
Anti-locality: To determine which phrasal projection can move to Spec, *-rare*, we adopt a version of anti-locality (see Abels 2003, Kayne 2005), which we informally formulate in (35):

(35) Anti-locality: Given H with a [+epp] feature, and XP its complement, XP may not I-merge with H:

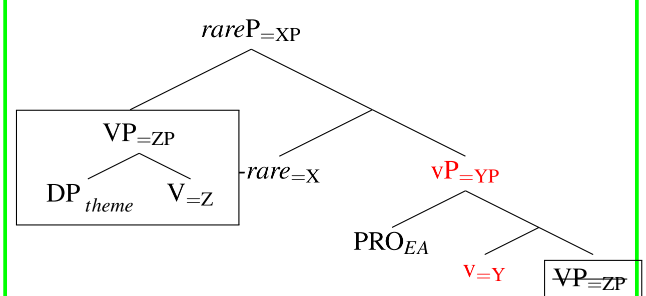
$$[\ H_{epp(+V)}\ [_{XP}\ ..\ X\ [_{YP}\ ..V..\ ]\ ]\ ]$$

Because of anti-locality, the immediate complement of the head is prohibited from I-merging to its Spec, as shown in (36). Instead, as shown for canonical (agent theme) passives in (37), the immediate complement of the next head down, i.e, ZP containing the V, is attracted to Spec, *-rare* under Attract Closest, stranding the silent external argument. Recall that we represent Japanese as head initial structures, except for the big VP and the PP. Head finality is derived by leftward phrasal movement to the large number of verbal suffixes.

(36) **Excluded:** movement of the immediate complement of X to the Spec,X.



(37) ✓ YP is frozen, ZP, the immediate complement of Y containing V I-merges with *-rare*.



Attract Closest: The movement of the big VP to Spec, *-rare* brings the theme closest to the nominative Case position. This is essentially the smuggling derivation of Collins (2005b).[38] This derivation gets around the classical Minimality problem (as the external argument does not intervene for the movement of the object).[39] It prevents the external argument from mapping to the nominative, in effect "demoting" it. It allows the external argument (a vol agent in the case of basic passives) to be projected in the syntax, simplifying the representation of thematic structure. It accounts for control into purposives, which require a vol agent, etc. Its silence can be accounted for independently, given the particular configuration of passive voice.

Crucially, what verbal constituent will move to Spec, *-rare* depends on the size of *-rare*'s complement, i.e. on the structural height at which *-rare* merges. It will, however, always be the immediate complement of the head selected for by *-rare*, i.e. ZP in (37), as this is the closest verbal phrase that can move. This implies that figuring out the fine derived syntactic structure of ZP in a particular language will play a crucial role.

[38] For general remnant movement analyses and their effects on building surface constituency, see Kayne's collected works (Kayne 1994, 2005, 2010, 2019). Many of Kayne's derivations are in fact smuggling derivations (see also §10.3). For generalized smuggling, see Collins (2005b,a) Belletti & Collins (2021), Angelopoulos et al. (2020), Koopman (2012b, 2021a, 2024) among others.

[39] This analysis differs from the standard unaccusative analysis in minimal ways. The smuggling derivation simply starts deeper in the structure. The output of the big VP movement is, in fact, the starting point of the derivation in the standard unaccusative analysis.

7.2 Derivations and discussion

In this section, we sketch the derivations for passives and psv-potentials of low obliques, showing that goal, directional, and source are indeed lower than little *v*. We then turn to the derivation of a locative psv-potential (i.e, high psv-potential), which suffices to make our point about a unified analysis.

7.2.1 Goal, directional, and source passives

Low obliques like directional '*to*' or source '*from*' are paths, which combine with a location describing the endpoint or starting point of the theme. The directional 'to' path construction has been analyzed as being low in the structure, viz, as being part of the extended big VP domain. (See Kishimoto (2008), Miyagawa & Tsujioka (2004) and Pylkkänen (2008) for Japanese, Hoekstra & Mulder (1990) for small clause analysis for Dutch directional PPs, Li & Thompson (1974) and Sybesma (2013) for Mandarin.) This is also true for source 'from'; for example, Pylkkänen (2008:p.68) has analyzed source with a transitive verb *nusum-u* 'to steal' as a low applicative, which is consistent with the current proposal.[40]

The goal in ditransitive/double object constructions is also structurally low, lower than the external argument. In the case of ditransitive constructions, what maps onto the nominative in the passive depends on the noun phrase that ends up closer to the nominative position: the goal or the theme. In Japanese, either can be promoted (Kishimoto 2001, Miyagawa & Tsujioka 2004, contra Kuroda 1992), except that the overt *ni* 'by'-phrase is not available when the theme is promoted to the nominative (see Ishizuka 2010, 2012, Jo & Seo 2023). Though the structure of Japanese ditransitive/double object constructions remains a major topic in the literature, it is uncontroversial, given the passivization pattern, to treat the goal, even the high/possessor goal in the sense of Miyagawa & Tsujioka (2004) or Kishimoto (2004), as occurring under the agent introducing little *v*, within the (extended) domain of big VP (see Ishizuka 2019, Yatsushiro 2003, among many others). Therefore, goal and directional '*to*' & source '*from*' count as low obliques, which are in the c-command domain of the ${}^{Low}\text{Pass}_{rare}$ and can undergo (low and regular) passivization (see (27)).

7.2.2 Derivation of goal, directional, source passives (cf. Ishizuka 2010, 2012)

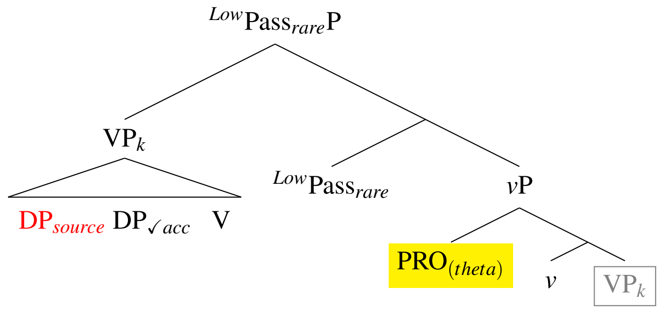
Passive Voice *-(r)are* is a phrasal suffix, expressed with an epp[+V] feature that needs to be satisfied by merging a phrase with a verb to its Spec. Due to anti-locality (Abels 2003, Kayne 2005), (see §7.1), the immediate complement of *-rare*, i.e. *v*P, cannot move to Spec, *-rare*, but the complement of v, big VP, is, the closest phrase that can. Since low obliques are in the big VP domain, VP movement to Spec, *-rare* will bring the low oblique closest to the nominative position, as illustrated in (38).[41]

(38) a. Sono ginko-ga (kokyaku-no) okane-o nusum -are -ta.
that bank-NOM (client-GEN) money-ACC steal -(R)ARE -PAST
Lit 'That bank was stolen (clients') money from.'

[40] Source PPs can in fact be low when expressing the source of the theme, but also high, locating the external argument, as shown in examples like *they observed the fire* [<*they*> *from the roof*] (cf. Ed Keenan, pers.comm). See also Pantcheva (2011) on the differences of source vs goal directionals.

[41] We indicate (checked) Case of a DP by ✓ (e.g, $\text{DP}_{\checkmark acc}$), and lack of Case with the absence of ✓ and in red (e.g, DP_{source}).

b.



Oblique source nominatives are compatible with a descriptive reading. We take this to show that they move to the low nominative *-ga*, merged in the clausal spine above PassP, yielding a regular passive. As stated in (9), we assume that oblique DPs are Caseless if they lack the structural layer where overt P(ostpositions) appear. Hence, oblique DPs move to the low nominative *-ga* for Case reasons, i.e. they undergo A-movement.

7.2.3 Derivation of goal, directional, source low psv-potentials

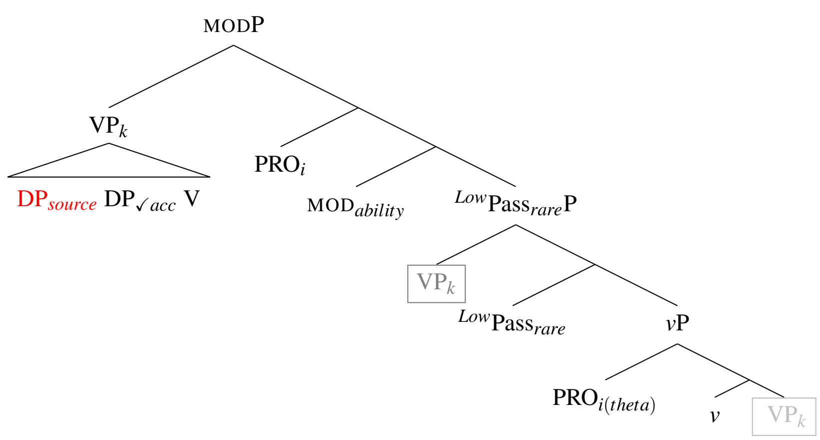
Goal, directional and source psv-potentials are built on the ^{Low}Pass$_{rare}$ structure in (38-b) by merging MOD$_{ability}$. We sketch (a simplified) derivation for (39) (which is structurally ambiguous between an psv- and active potential reading) focusing on the relevant pieces of the derivation.

(39) Sono ginko-ga (kokyaku-no) okane-o nusum -e -ru.
that bank-NOM (client-GEN) money-ACC steal -(RAR)E -PRES
Lit. 'That bank can be stolen (the clients') money from.

We analyze MOD$_{ability}$ as a control predicate, projecting a subject that controls the external argument of the main predicate. Since the silent subject of MOD$_{ability}$ cannot map onto nominative *-ga*, and alternates with a *ni* phrase, we tentatively assume it is PRO (rather than *pro*). Furthermore, since the low oblique maps onto nominative *ga*, we conclude that the VP in Spec, *-rare* must have moved to a position above PRO,[42] from where it can map onto the high nominative. This yields the linear order in (39), with V preceding *-rare*, as in (39).

(40) a. .. ga$_{high}$ [$_{VP}$ DP$_{oblique}$... V]$_k$ [PRO$_i$ MOD$_{ability}$ [[$_{VP}$ DP$_{oblique}$... V]$_k$ rare [$_{vP}$ PRO$_i$ v VP]]]]

b.



[42]This position should be involved in licensing PRO.

As we can see, extraction of the oblique happens at a much higher point in the derivation than extraction of obliques in the regular passive which takes place from Spec, *-rare*. This yields insight into why obliques in psv-potentials, but not obliques in passives, must map onto the high exhaustive *-ga* in root clauses.

As we discussed in §4.2 (see (22) and also §5), c-command by *-rare* is a necessary condition for the mapping onto the nominative. Thus, obliques below little *v* are possible candidates for passivization. In contrast, high obliques, like locatives, instrumentals and manners, are not, since they are merged above *v*P and passive Voice. This accounts for part of (28-b): high obliques are excluded in the passive construction in Japanese, but low obliques are possible.

7.2.4 Locative and instrumental high psv-potentials

We now turn to high psv-potentials with high obliques (e.g, locative, instrumental) and test our hypothesis for a unified analysis of passive Voice *-rare*. Note that the locative and instrumental high psv-potentials in (41) include the perfect/progressive *-tei*.

(41) a. Kono kafe-ga (manga -o yomi-nigara) keeki-o tabe-tei -rare-ru.
this cafe-NOM (comic.book -ACC read-while) cake-ACC eat-PERF/PROGR -RARE-PRS
Lit. 'This cafe can be eating cake in (while reading comic books).'
≈ It is possible to stay in this cafe having eaten a cake while reading comic books. Locative (*de*)

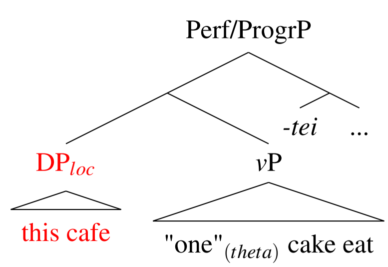
b. Kono pen-ga (nagai-aida) kai -tei -rare-ru.
this pen-NOM (long-period) write -PERF/PROGR -RARE-PRS
Lit. 'This pen can be writing with (for a long time).'
≈ It is possible to keep writing with this pen for a long time. Instrumental (*de*)

We present the derivation of (41-a) without a *nagara* (while) clause in several steps below, in §7.2.5–§7.2.8.

7.2.5 Derivation (*part 1*)

We start the derivation at the point where perfect/progressive *-tei* has been merged into the structure. In (42), the projection containing the locative has pied-piped the *v*P to the left of the phrasal suffix *-tei*, which, like *-rare*, requires a verbal specifier.[43]

(42)



As (42) shows, we assume that the external argument is merged in Spec, *v*P, below the locative adjunct. This is crucial for our analysis, as we will further discuss in the next section and in §11.2. (We return to the question how the external argument starts out below the locative or instrumental DP/PP, but ends up occupying the nominative *-ga* position in the active in §11.2.)

[43] The perf/progr marker *-tei* appears to be bimorphemic, consisting of *-te* and *-i* (see Nakatani 2004, among others). Hence this movement to Spec, *-tei* is compatible with anti-locality.

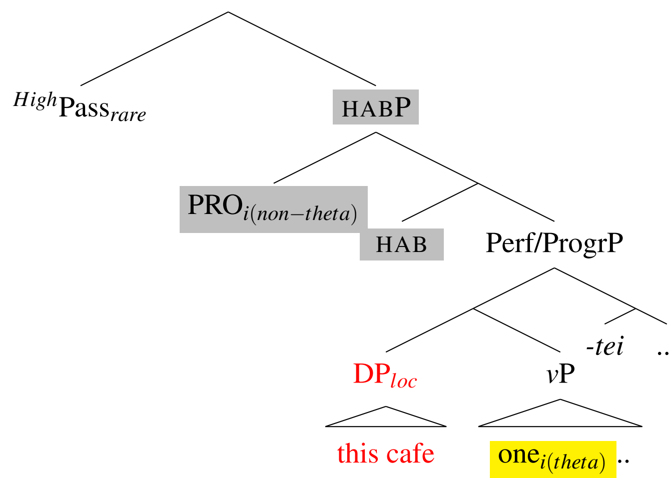
7.2.6 Derivation (*continued, part 2*): On the independence of passive voice and argument structure

In the next steps of the derivation, *-(rar)e* merges with the structure (42), which must be bigger than Perf/progrP. As shown in (31), the perfect/progressive *-tei* precedes the suffix *-rare* ([[V-*tei*]-*rare*]-(*ru*)) in high psv-potentials, thus given anti-symmetry, anti-locality, and attract closest, the complement of *-rare* must contain *-tei*, embedded under exactly one other projection H. Taking our inspiration from Cinque's hierarchy, where Habitual is immediately below *possibility*, we suggest that H is the (silent) habitual (i.e. HAB) with (arbitrary/generic) PRO in its Spec.

We thus suggest H is in fact a projection of the habitual HAB, which is involved in the interpretation of the silent generic pronoun.[44] This leads us to the complement structure illustrated in (43).

Note that this structure is quite comparable in size to the structure of the infinitival complement of *possible* in English: it basically captures the bi-clausal nature of the alethic *possibility* interpretation.

(43) Merge Passive *-(rar)e* with HABP:



First, since the perfect/progressive *-tei* is a prototypical raising predicate, not selecting for an external argument, the subject position in the complement of *-rare* must be analyzed as a *derived, non-thematic* subject position. As stated in §7.2.5, the thematic position of the external argument must be located *below* the perfect/progressive and locative or instrumental DP/PP (i.e, Spec, *v*P).

Furthermore, our analysis forces a silent PRO under *-rare* but *above* the locative oblique, as it can bind into instrumental (and locative) PPs (see §8.2). PRO cannot be inside the Perf/ProgrP: if it were, it would be the highest argument in the attracted Perf/ProgrP, and block promotion of the locative oblique to Spec, *-ga*. We therefore concluded that it must be *outside* of the projection of Perf/Progr *-tei*. This is why we located it in HABP.

This reduces the silence of the external argument to the well-known distribution of PRO in infinitivals. Basically *-rare* "imposes" silence on a (non-thematic) subject position, for the same reason the infinitival complement of '*possible*' in English contains a silent arbitrary PRO. High psv-potentials like (43) are important since they show that **the obligatory silence of PRO cannot be related to the argument structure of the main predicate**, but must be due to the syntactic principle that regulates where exactly PRO can

[44] This seems to hold crosslinguistically, regardless of whether it is 'silent,' as for impersonal *si* in Italian (Cinque 1988), the impersonal passive in Turkish (Legate et al. 2020) or the accusative passive in Maasai (Greenberg 1959), or 'overt,' as for German *man*, Dutch *men*, French *on*, English *one* or *they* in many African languages.

survive. Thus:

(44) a. An external argument can be silenced under *-rare*, regardless of whether it is in a thematic position or not, and

b. The presence of an external argument in the syntactic representation is compatible with the presence of Passive Voice.

Japanese high psv-potentials, thus, provide a way to distinguish different proposals on the nature of passive voice, as the external argument must be present in the syntactic structure, and the silence of an external argument is independent of the thematic structure.

We now also have a hypothesis as to what the immediate complement of high modality dependent *-rare* is. Namely, *-rare* merges with a subject-predicate structure headed by the aspectual category Habitual. It contains a silent non-thematic A-position, comparable to an infinitival complement in English. (45) illustrates the complement structure for high psv-potentials (45-a) and for passive/low psv-potentials (45-b).

(45) Complement of *-rare* in high and low psv-potentials:

a. ✓[-rare [$_{\text{HABP}}$ PRO$_{-theta}$ HAB (Perf) (Progr) Pred]] $\boxed{\text{High psv-poten}}$

b. ✓[-rare [$_{v\text{P}}$ PRO$_{+theta}$ v VP]] $\boxed{\text{Passive/Low psv-poten}}$

These are specific cases of the more general configuration (46) in Japanese, with the DP in subject position silent, regardless of whether it is in a theta-position or not:[45]

(46) Passive voice *-rare* can merge with: (*final version*)
✓ [-rare [PRO$_{\text{(non)thematic}}$ H Pred]], where H is little *v* or Habitual, and Spec, HP (PRO) is silent.[46]

Arguments supporting the presence of a silent element in Spec,HabP are discussed in §8.2. In §9, we show (46) also extends to active potentials in Japanese.

Our analysis raises the question if the external argument is introduced in the *v*P (or alternatively for Voice affectionados, (active) Voice) *below* the locative, as we propose, or *above* the locative (as generally assumed in the (active) Voice literature, following Pylkkänen (2002, 2008)). If the latter, however, we lose our account for why high obliques like locatives or instrumentals cannot be passivized in the regular passive. We support our analysis with Japanese internal evidence in §8, and with crosslinguistic evidence from Malagasy (Austronesian) in §11.1.1 and ciCewa (Bantu) in §11.1.2.

7.2.7 Derivation (*continued, part 3)*

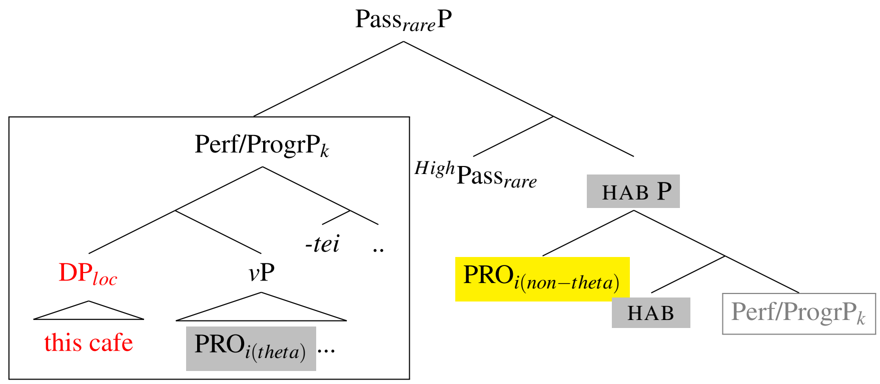
The next steps of the derivation of (43) are given in (47), deriving the linear order [Loc... V-*tei-rare*], characteristic of the high psv-potential.

(47) I-merge the complement of Habitual, $\boxed{\text{Perf/ProgrP}}$, with Spec, *-rare* to satisfy the lexical property

[45] Note (46) potentially applies to long passives in Japanese (Ariji 2006, Ishii 2012, Ishizuka 2010, 2012, Sugioka 1984, Nishigauchi 1993, Wurmbrand 2012, and others), where the passive morpheme does not follow the lexical verb, but a bound aspectual, control, or raising restructuring predicate. See Appendix C.

[46] As stated in 36, a *ni* 'by'-phrase is available for passive and low psv-potentials, but not for high psv potentials. We assume *ni* is merged in a region above passive voice and the ability MOD.

of *-rare*.

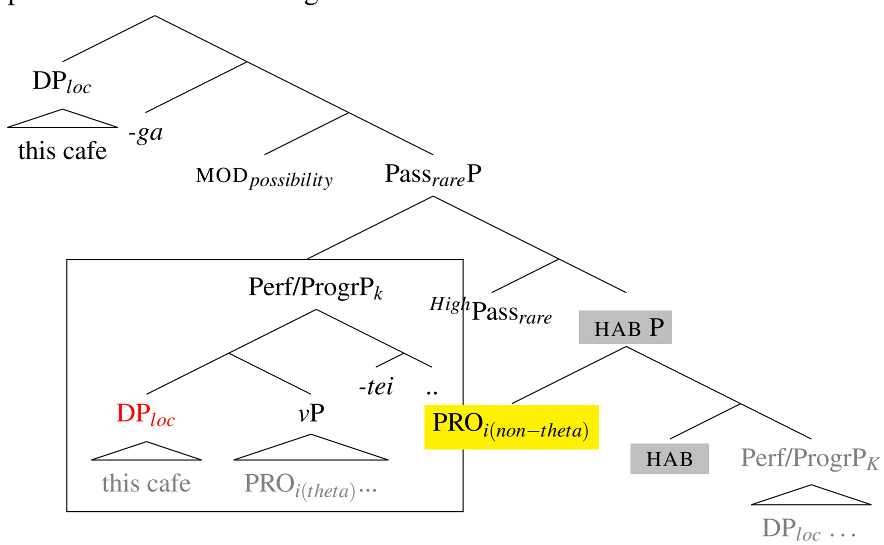


7.2.8 Derivation (*continued, final steps*)

The final steps in the derivation involve the merger of the MOD$_{possibility}$, merge of T, *-ga*, and I-merge of the closest Caseless locative DP, *this cafe*, to Spec, *-ga*. These steps are illustrated in the structure below:

(48) Merge MOD$_{possibility}$, merge T, *-ga*, I-merge locative with *-ga* (i.e. Attract Closest):

Kono kafe-ga keeki-o tabe -tei -rare (-ru)
this cafe-NOM cake-ACC eat -PERF/PROGR -RARE (-PRS)
Lit. 'It is possible to continue eating cake in this cafe.'



8 Further support

8.1 The hierarchy of high obliques: Locative> Instrumental> Manner

So far we identified locative, instrumental, and manner as high obliques that can only occur in the high psv-potential. This is because they are merged higher than little *v* and are not c-commanded by ^{Low}passive voice *-rare*. We next consider the question if these obliques are rigidly ordered, and if so, how this order relates to the proposed universal hierarchies. In his discussion of the order of temporal, locative, and manner PPs, Cinque (2009, 2004: ch.6), citing Schweikert (2005), shows that Temp>Loc>Manner appear to be rigidly ordered in languages where they precede the V (controlling for focus). In contrast, when they follow the V, a wider variety of orders appear to be attested, yet certain orders are unattested. Importantly, these possible and unattested orders are the same type as found in Greenberg's Universal 20 (Cinque 2005, 2023).

There are two arguments for a Temp> Loc>Instr/Manner hierarchy in Japanese. The first one comes from the unmarked ordering of PPs. It basically corresponds to the preverbal universal order, except that the relative ordering between instrument and manner is unclear: it at least gives us Temp>Loc> Instr/Manner.

(49) Kinoo kooen-de {kureyon-de /sono hohoo-de} e -o kai -ta.[47]
yesterday park-LOC {crayon-with /that method-with} picture -ACC draw -PAST
'Yesterday (I) drew a picture {with crayons/with that method} in the park.'

The second argument is based on Minimality effects that are generally attributed to 'Attract closest.' Given Attract closest, we expect that if two or more obliques/PPs are present, only the highest is promotable, since extracting the lower one yields a Minimality Violation.[48] A pairwise comparison leads to the following hierarchy Locative>Instrumental> Manner, suggesting the high obliques that are possible in psv-potentials are ordered in a stable (and most likely) universal hierarchy. This is illustrated below:

(50) Locative > Instrumental
a. Kono zimu-ga atarasii masiin-de karada-o kitae-rare-ru.
this gym-NOM new machine-with body-ACC train-RARE-PRS
Lit. 'This gym can be trained (your) body in with new machines.'
≈ 'It is possible to train (your) body with new machines in this gym.'
b. Atarasii masiin-ga (*kono zimu-de) karada-o kitae-rare-ru.
New machine-NOM (*this gym-LOC) body-ACC train-RARE-PRS
Lit. 'New machines can be trained (your) body with (*in this gym).'
≈ 'It is possible to train (your) body with new machines (*in this gym).''

(51) Locative > Manner
a. Kono zimu-ga sono hohoo-de karada-o kitae-rare-ru.
this gym-NOM that method-with body-ACC train-RARE-PRS
Lit. 'This gym can be trained (your) body in with that method.'
≈ It is possible to train (your) body with that method in this gym.'
b. Sono hohoo-ga (*kono zimu-de) karada-o kitae-rare-ru.
that method-NOM this gym-LOC body-ACC train-RARE-PRS
Lit. 'That method can be trained (your) body with (*in this gym).'
≈ 'It is possible to train (your) body with that method (*in this gym).'

(52) Instrumental > Manner
a. Kono masiin-ga sono hoohoo-de karada-o kitae-rare-ru.
this machine-NOM that method-with (body-ACC) train-RARE-PRS
'This machine can be trained (your) body with with this method.'
≈ 'It is possible to train (your) body with that method with this machine.'

[47] Switching the order between instrument and manner in (49) does not degrade the sentence. Definiteness may play some role: *sono hohoo-de* 'that method-with' is definite.

[48] Or more precisely, given our assumption that oblique DPs are smaller than full PPs, it must be due to the hierarchy of merge that governs the relative merge order of Ps to form full PPs: Indeed, only the highest oblique can be a DP in a string of obliques. All other obliques are forced to be full PPs.

b. Sono hohoo-ga (*kono masiin-de) karada-o kitae-rare-ru.
that method-NOM (*this machine-with) body-ACC train-RARE-PRS
Lit. 'That method can be trained (one's) body with (*with this machine).'
≈ 'It is possible to train (your) body with that method (*with this machine).'

The examples above not only support the universal hierarchical order among the high obliques, but also provide strong evidence for the movement analysis of psv-potentials in Japanese.

Finally, we provide language-internal support for the two subject positions from a string of indefinites, one higher and one lower than locative adjuncts (see (71-a)). As shown in (53), an indefinite subject must follow an indefinite temporal and can precede or (slightly less natural) follow an indefinite locative.

(53) a. Itu-ka doko.ka-de dare.ka-ga kimi-o mituke-ru-daroo.
'when'.Q 'where'.Q-LOC 'who'.Q-NOM you-ACC find-PRS-will — Temp>Loc>Sub

b. Itu.ka dare.ka-ga doko.ka-de kimi-o mituke-ru-daroo.
'when'.Q 'who'.Q-NOM 'where'.Q-LOC you-ACC find-PRS-will
'(I think) someone will find you somewhere sometime.' — Temp>Sub>Loc

We take this to show there is a subject position below temporals, but **higher** or **lower** than a locative, i.e. Temp_{ind}>$(\text{Sub})_{ind}$>Locative>Sub_{ind}. This is, therefore, compatible with our analysis in (71-a), with a non-thematic A-position above the locative or instrumental, and a thematic position below these.

8.2 Binding and Control

We have analyzed the silent implicit argument as a PRO in Spec, HABP for high 'possibility' psv-potentials. Here we present two further arguments that confirm this high position. The first is based on binding. PRO can bind an anaphor in high oblique PPs like instrumentals, as illustrated in (54).

(54) Kono kafe-ga PRO_i [**zibun(.zisin)$_i$-no** kappu-de] kohi-o nom **-e** -ru.
this cafe-NOM PRO_i [self(.self)$_i$-GEN cup-with] coffee-ACC drink -(RAR)E -PRS
Lit. ✓ 'This cafe can be drunk coffee in with self's$_i$ cup (by one$_i$.)'
≈ 'It is possible for one$_i$ to drink coffee with one's$_i$ cup in this cafe.' — High psv-potential

Note that *zibun* is a logophor and a long-distance anaphor, generally glossed as monomorphemic *self*, but decomposable into *zi.bun* (=self.part). *Zibun.zisin* =[self.part].[self.body] is a local anaphor (and an emphatic reflexive)(see Katada 1991, Ikawa 2024, among others). (54) is well formed with either *zibun* or *zibun.zisin*.

Secondly, the implicit argument in the psv-potential controls a PRO in a temporal (simultaneous) adjunct clause headed by *-nagara* 'while.' This is a typical diagnostic for subjecthood, as argued by Ura (1999:p.230) and Kanno (1992), among others.

(55) a. [PRO_i (Yuka-ni) ne.korobi-nagara] kono tosyokan-ga PRO_i hon-o yom -e -ru.
(floor-DAT) lie.down-while this library-NOM book-ACC read -(RAR)E -PRS
Lit. ✓ 'This library can be read books in while lying down (on the floor).' — loc. pass
≈It is possible (for one) to read books in this library while lying down (on the floor).'

b. [PRO_i Ne.korobi-nagara] manga -ga PRO_i sono tosyokan-de yom-dei -rare -ru.
lie.down-while comic.book -NOM that library-LOC read-ASP -RARE -PRS
Lit.✓ Comic books can be read (by one$_i$) in that library while (one$_i$ is) lying down.' — theme pass

≈'It is possible (for one$_i$) to be reading comic books in that library while (one$_i$) is lying down.'

The nominative subjects of (55-a) and (55-b) are locative and theme, respectively. Note that the linear order *yom-tei-rare*, with perf/progressive *tei* as inner suffix, insures that this is indeed the high psv-potential.

So far, we have shown that the implicit argument in the high psv-potential can bind and control. The question is if it in turn can be controlled by an outside controller, as Legate et al. (2020) show for Turkish. We have so far been unable to construct cases similar to Turkish. However, this could be explained by the very high position of PRO in the clause: the silent generic subject (PRO) in Spec, Habitual under MOD$_{possibility}$ is too high to be c-commanded by *want* for example (see (88) and (93)). It thus cannot be controlled by the subject of *want*.[49] However, a hanging Topic can determine the reference of the PRO, as shown in (56).

(56) Ken -ni -tuite -iwe -ba, sono mise-ga nagai-aida hon-o yom -dei -rare -ru.
Ken -DAT -about -say -if that shop-NOM long-period book-ACC read -ASP -RARE -PRS
'Speaking of Ken, that shop (he) can continue reading books (in) for a long time.'
≈ Speaking about Ken, he can continue reading books in that shop for a long time.'

We conclude from the above that there clearly is a syntactically projected external argument in the high psv-potentials. Nevertheless, what about the external argument of a run of the mill (vol) agent-theme passive? Is it also syntactically projected or not, as Legate et al. (2020) propose? While the silent external argument, as is well known, can control, it fails to be able to be controlled.[50] The question is if this failure can be explained by independent principles, or if it can be taken as evidence that the implicit argument is absent in the syntax. We note that in our analysis, there always is some argument c-commanding the silent subject embedded under the (basic) passive Voice. If this argument is always c-commanding PRO, a higher controller could could be blocked for that reason (i.e. *DP$_j$ .. [DP$_i$ ga [[$_{VP}$ t$_i$ V][rare [PRO$_j$ *v* ~~VP~~]).[51]

The promoted theme argument itself, however, cannot bind PRO$_{agent}$ in basic passives either, as this should be reflected in a possible reflexive reading. This is excluded in Japanese, and, as far as we know, crosslinguistically, even in languages like Turkish, where a Middle Voice homophonous to the passive, can lead to a reflexive reading (see Key 2024, for discussion).[52]

(57) Zibun(zisin) -ga araw -are -ta.
self(self) -NOM wash -(R)ARE -PAST
'I was washed (PRO=by someone else), *by myself.'

We suggest here that this is due to the timing(s) of the Binding principles. Since a (vol) agent determines a complete functional complex, *v*P counts as a binding domain. It is therefore expected that Binding theory applies before VP movement to Spec, *-rare* takes place. As long as the theme does not refer to the agent, and is a pronoun or an R-expression, either Principle B or C will yield the desired results. However, what would

[49] The subject of *ta.i/gar* 'want' resists being controlled by the generic PRO.

[50] However, see Koopman (2024, 2.7.1) for an argument that the very low implicit argument in Mandarin passive-resultatives, in fact, *must* be controlled by the surface object.

[51] Japanese does not have impersonal passives. However, even in languages with impersonal passives there might be a c-commanding expletive, depending on one's analysis of expletives −still a debated theoretical issue.

[52] Key (2024) follows others and stipulates that passive voice (but not middle voice) comes with a presupposition that the agent and theme must be disjoint. Note that Japanese *rare* also occurs in the middle construction, but never yields a reflexive reading. This is something which begs for a further explanation.

rule out a derivation where the theme starts out as a reflexive bound on the *v*P cycle? After VP movement to Spec, *-rare*, and movement of the theme to the nominative subject position (an A-position), the nominative anaphor would not be expected to violate Principle A at the level of the TP. So, perhaps what should be ruled out is for *zibun* to start out as the theme object under the passive. If *zibun* has to start out above the Passive, it would be ruled out on the TP cycle.

(57) is good, however, under a logophoric reading of *zibun* (provided *zisin* is absent), bound on the CP level. However, even in this case, PRO has to be disjoint from the logophor. This now looks like an effect of principle B, with the logophor (like logophoric pronouns) subject to Principle B on the *v*P cycle.

While a short passive with PRO does not permit a reflexive reading (see (57)), a *ni* by-phrase containing an overt reflexive anaphor is in fact OK:

(58) Ken-ga zibun.(zisin) -ni sukuw -are -ta.
Ken-NOM self.(self) -DAT save -(R)ARE -PAST
'Ken was saved by himself' or 'Ken was saved by speaker (*cf: only available for bare zibun*).

We take this to show that the anaphor in the *ni*-phrase is bound by the nominative subject on the TP cycle, and therefore that the *ni* by-phrase must be outside the little *v*P. This implies that the *ni* by-phrase and the silent external argument are not in the same syntactic position (contra Collins 2024, see also fn 36).[53]

We thus suggest that the disjointness of a volitional agent and a theme in the canonical passive might be derivable from the syntactic structure and the timing of the principles of Binding Theory. It is precisely the fact that the transitive *v*P structure is merged below passive Voice, which allows exploring this possibility.

Why can the external argument in passive constructions always be silent? We saw in high psv-potentials that the external argument of the thematic verb must be silent, but that this cannot be due to a property of *-rare* resulting in the reduction of the argument structure of the lexical verb. In the case of high psv-potentials, *-rare* is simply nowhere close to the main predicate, which is deeply embedded in *-rare*'s complement. Yet, the complement contains a generic PRO located in a high (non thematic) subject position, forming a chain with the external argument (or the highest argument) of the predicate in the

[53]Nakamura & Fujita (1998) argue against a passive analysis for (psv)-potentials (see fn 3). As they show, the derived (theme) subject in passive and potential constructions behaves differently w.r.t. binding of subject-oriented *zibun*. We reproduce their contrast for passive and psv-potentials here with somewhat simpler examples:

(i) a. Ken$_i$-ga Naomi$_j$-ni zibun$_{i/*j}$-no heya-de nagur-are-ta.
Ken-NOM Naomi-DAT self-GEN room-LOC hit-PASS-PAST
Ken$_i$ was hit by Naomi in self$_{i/*j}$'s room. | Passive: Ken-ga = self |

b. Ken$_i$-ga Naomi$_j$-ni zibun$_{*i/j}$-no heya-de nagur-e-ru.
Ken-NOM Naomi-DAT self$_j$-GEN room-LOC hit-POT-PAST
Ken$_i$ can be hit by Naomi$_j$ in self$_{*i/j}$'s room. | Theme psv-potential: Naomi-ni=self |

In a nutshell, the theme *Ken* in the passive example in (i-a) can be the antecedent of subject oriented *zibun* inside the Locative PP, but the DP in the ni-phrase cannot. The latter can in the potential example in (i-b), but the nominative theme cannot.

Note that the fact there is a difference between (i-a) and (i-b) is not an argument that passive *-rare* is not involved in the potential. Indeed, for us the *ni* by-phrase in the passive is not c-commanding the locative PP in the passive in (i-a), if the locative is merged above the Passive phrase including the *ni* by-phrase. In (i-b), however, in agreement with Nakamura & Fujita (1998), the potential has an antecedent subject since *ability* is a control predicate, high enough to allow binding into the locative DP (which is merged below the Mod introducing *ability*, blocking binding by the (theme) nominative subject in (i-b). If so, the difference can be explained in independently motivated configurational terms. A full account will have to wait for future research. We do not have space here to discuss their second argument on honorific constructions: they fail to make their case because of minimal pair problems.

infinitival. We conclude, therefore, that the silence of the external argument should not be related to 'its absence' from the syntactic structure, but should be attributed to whatever general configurational principle of (non)pronunciation determines where PRO can or cannot appear.

8.3 Scope

The proposed derivation makes predictions in terms of scopal interactions of the modality and a quantified oblique nominative or PP. Nakamura & Fujita (1998, p.180, (20a) & (20b)) report that a nominative-marked instrument with *dake* 'only' (eg, *pen-dake-ga* 'pen-only-NOM') always takes wide scope with respect to the MOD, while an oblique-marked one (*pen-dake-de* 'pen-only-with') is ambiguous between a wide and narrow scope reading. This is the same scope pattern as found for the nominative and accusative Case alternation on objects in active potentials, with nominative with *dake* 'only' only scoping above (*-(rar)e*) (see Tada 1992, Koizumi 1998, Saito & Hoshi 1998, Takano 2003, Bobaljik & Wurmbrand 2007, among others).[54] We reproduce the same pattern for locatives in (59).

(59) Nominative-Oblique alternation of Locative

a. Tosyokan -dake -**ga** sono hon-o mituke -rare -ru $\boxed{\text{Locative Psv-potential}}$
library -only -NOM that book-ACC find -RARE -PRS
'One can find that book only in a library (but nowhere else).' (only>can, *can>only)

b. Tosyokan -dake -**de** sono hon ga PRO mituke -rare -ru. $\boxed{\text{Theme Psv-potential}}$
library -only -LOC that book -NOM find -RARE -PRS
'One can find that book only in a library (no need to go anywhere else).' (only>can, can>only)

Our analysis captures these basic scope patterns. We box the relevant elements for scope in the structures we have motivated below:

(60) a. DP_{loc} (only) $\boxed{\textbf{ga}}$ [$\boxed{\text{MOD}_{possibility}}$ Pass_{rare} > ...DP_{loc}... =(59-a)

b. $\boxed{\text{[DP (only) } \textbf{de} \text{]}_{PP}}$ DP ga [$\boxed{\text{MOD}_{possibility}}$ Pass_{rare} [$\boxed{\text{DP } \boxed{\textbf{de}} \text{]}_{PP}}$ ....]] =(59-b)

The relevant property seems to depend on the level at which the nominative Case and Case assigning Ps are merged. In (60-a), the nominative case marker *-ga* is merged only *above possibility*, whereas the P *de* in (60-b) is merged *below possibility*, with the PP subsequently A'-moved *above possibility* (and the theme moved to *-ga*). Thus the *ga*-marked oblique is in an A position in (60-a), and cannot reconstruct below *possible*. In contrast, the A'-moved PP in (60-b) *can* reconstruct to the position under *possible* where the PP is formed, or take surface scope.

This is consistent with our assumptions (i) that oblique DPs move to an A-position where they get Case marked by *-ga*; (ii) that (the high) *-ga* is merged above high potential *-(rar)e*; and (iii) that the minimal scope is determined by the Case position of the DP.

This concludes the empirical and analytical discussion of psv-potentials and passives. We have motivated an unified analysis for *-rare*: it looks like a passive, behaves like a passive, and therefore it is the same

[54]The literature makes different claims about whether the scope of the accusative *must* be below *-(rar)e*, or whether it can also be above *-(rar)e*. The first author agrees that such sentences are ambiguous in the appropriate context, with the accusative scoping below or above *-(rar)e*.

passive. Differences reduce to the different heights at which *-rare* in psv-potentials and passives merge.

9 Extending the unified analysis to Active Potentials

So far we have shown that the same passive voice *-rare* occurs in both passives and psv-potentials. We now turn to active potentials, and ask:

(61) Can the passive voice analysis we developed be extended to derive Active Potentials?

If it can, it will be a strong confirmation that our analysis is on the right track. According to our hypothesis, the difference between active and pvs-potentials could only come about by differences in *rare*'s local environments, which includes the relative order of merge.

Active potentials look like active sentences: the external argument, which can be an agent, a natural force, or an inanimate, maps on to the nominative subject, as exemplified below:

(62) a. Mei-ga ninzin-o tabe -rare -ru.
Mei-NOM carrot-ACC eat -RARE -PRS
'Mei can eat carrots.' $\boxed{\text{Agent}}$

b. ?Sono taihu -ga kono yane-o hukitobas -e -ru -kana.
?that typhoon -NOM this roof-ACC blow.off -(RAR)E -PRS -wonder
'(I wonder if) it is possible for that typhoon to blow off this roof.' $\boxed{\text{Natural force}}$

c. Sono kuni -ga imin -o uke.ire -rare -ru.
that country -NOM immigrant -ACC take.in -RARE -PRS
'That country can take in immigrants.' $\boxed{\text{Inanimate subj}}$

Removal of the potential *-rare* in examples in (62) leads to well-formed non-modal active sentences, so it just looks like *-rare* is a simple modal head, which is how the Japanese literature treats it: it is generally glossed as 'can.' This raises the question why there is a passive morpheme in an otherwise active sentence—a property that recalls **deponent** verbs, a class of verbs with passive morphology, but active alignment (see Grestenberger 2014, among many others).

However, there are several arguments that the same *-(rar)e* occurs in both psv- and active potentials. As far as we have been able to tell, active potentials have exactly the same form and spell-out conditions on *-(rar)e* as psv-potentials. Active potentials take the suppletive form *deki-ru* in the noun light-verb construction in exactly the same contexts as psv-potentials (see appendix A). This strongly suggests that both MOD and passive Voice *-rare* are an integral part of the structure of potentials, regardless of whether they are psv- or active potentials. Furthermore, active potentials have the same two modal interpretations as psv-potentials—'ability' and 'possibility'— with the former higher and the latter lower than aspectual *-tei*, as illustrated below:

(63) a. Ken-ga imadani sono huku -o ki **-rare** -tei -ru. $\boxed{tei > \text{Poten}}$
Ken-NOM even.now that clothes -ACC wear -RARE -ASP -PRS
Lit ✓'Ken can continue wearing those clothes even now.' [Low active-potential]
≈ 'It continues to be the case that Ken can wear those clothes now'
although those clothes look too small for him, he can still wear them

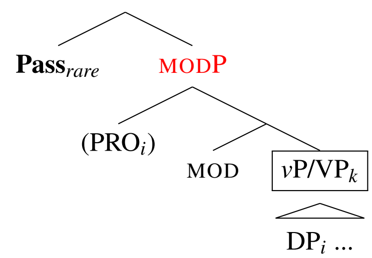
b. Ken-ga nannichi-mo sono huku -o ki -tei **-rare** -ru. $\boxed{\text{Poten} > \textit{tei}}$
Ken-NOM days-even that clothes -ACC wear -ASP -RARE -PRS
Lit. ✓‘Ken can be wearing those clothes for days.’ [High active-potential]
≈ ‘It is possible for Ken to continue wearing those clothes for (many) days.
even though they are filthy, he doesn’t seem to care.’

As we argued in §3, both heads—a silent potential modality and the passive voice *-rare*— have to be structurally adjacent to insure the correct spell-out form. As we pointed out there, this condition can be satisfied in exactly two configurations, either passive Voice is merged prior to MOD, as in psv-potentials (64) or MOD is merged prior to passive Voice, as in (64)ii.

(64) (i) MOD > Passive (cf. English *readable=can be read*), or $\boxed{\text{(ii) Passive > MOD}}$

We now show how (64)ii straightforwardly derives active potentials, without any further changes. (See Appendix D for a more detailed derivation.)

(64)(ii) Passive > MOD ⇒ Active potentials



Passive$_{rare}$ attracts a phrasal constituent containing a verb. MODP is frozen because of anti-locality. The immediate complement of MOD, $\boxed{v\text{P/VP/AspP}}$ is the closest verbal projection that can be attracted to Spec, Pass$_{rare}$, to satisfy its epp[+V] feature, and derive its suffixal status. This step will bring the highest argument *v*P (or VP) or AspP, for that matter, closest to the nominative. The output of the derivation will be a syntactically active structure, since the complement of Mod *v*P/VP has the alignment of an active sentence.[55]

Could potential support for the merge order Pass> MOD > *v*P/VP come from different sensitivity to verb class than Passives? (65-a) illustrates that regular passives cannot take an unaccusative VP complement. MOD in the active potential can, however (65-b). But so can psv-potentials (65-c), (which have a merge order MOD > Pass> ..*v*P/VP). This shows the distinction in fact between passive on the one hand and psv- or active potentials on the other.

(65) a. *Sono eki -ga Ken-ni (hitoride) ik -are -ta.
That station -NOM Ken-DAT (alone) go -(R)ARE -PAST
Int. * ‘That station was gone to by Ken (alone).’ $\boxed{\text{Passive}}$

b. ✓Ken-ga (hitoride) sono eki -ni ik -(ar)e -ta.
Ken-NOM (alone) that station -DAT go -(R)(AR)E -PAST
‘Ken was able to go to that station (alone).’ $\boxed{\text{Active potential}}$

c. ✓Sono eki-ga Ken-ni (hitoride) ik -(ar)e -ta.
That station-NOM Ken-DAT (alone) go -(R)(AR)E -PAST

[55] This type of analysis is suggestive of an analysis for so-called deponent verbs, in particular for those verbs that have passive morphology, but an (vol) agent-theme thematic structure. See also Koopman (2012a) for how an (ergative) active in Samoan derives from double passivization.

'That station could be gone to by Ken (alone).' Psv-potential

Furthermore, it turns out that not all unaccusatives behave in the same way. Inoue (2004) argues there are two types of unaccusatives that are distinguished by whether their argument is capable of volition ("control") or not. These data are summarized in the next table, with +/- volition distinguishing the unaccusatives.[56]

	verb		v-pass(-pst)	v-psv.pot(-prs)	v-act.pot(-prs)
1. unacc (-vol)	oti(-ru)	fall (from cliff)	*oti-rare(-ta)	*oti-rare(-ru)	*oti-rare(-ru)
2. unacc (+/-vol)	ik(-u)	go	* ik-are(-ta)	✓ ik-e(-ru)	✓ ik-e(-ru)
unacc (+/-vol)	korob(-u)	fall (on floor)	* korob-are(-ta)	✓korob-e(-ru)	✓korob-e(-ru)[57]
unacc (+/-vol)	tuk(-u)	arrive	*tuk-are(-ta)	✓ tuk-e(-ru)	✓ tuk-e(-ru)
3. (bare) unerg	oyog(-u)	swim	*oyog-are(-ta)	✓ oyog-e(-ru)	✓ oyog-e(-ru)

In sum, (i) potentials can combine with a broader class of predicates than passives,[58] and (ii) *oti-ru* 'fall' is incompatible with (ability) potentials for the same reason as control fails in **I am able PRO to rain*. Since passive *-rare* co-occurs with MOD, the location where MOD can merge determines where *-rare* is merged, and as we noted earlier, it might be there are 3 possible merge locations for *-rare* (cf. Appendix C).

One might wonder if the fact that passive Voice *-rare* in Japanese can take a clausal-like MODP complement is problematic or not. This is certainly not allowed in English, a generalization known as Visser's generalization ('subject control verbs do not passivize'). This difference might be related to the fact that Japanese has *long passives*, which English lacks. Long passives feature a mix of raising or control predicates.[59] Long passives are possible both with regular passives, as well as with (psv-) and active potential constructions (cf. Appendix C). A final difference is that English has no active potentials with *-able*, neither do the languages that we are familiar with that use passive or middle morphology for psv-potentials. This suggests Japanese homophony of active and passive potentials is probably rare crosslinguistically. Deriving active potentials from psv-potentials in the way we propose expresses both the historical relation of the passive and the psv-potential (a silent MOD + passive in a local configuration), and locates the core difference between psv- and active potentials exactly. According to Kanno (1992) the potential and passive forms have been tracking each other for many centuries, suggesting it might be a remarkable stable feature in the history of Japanese. Perhaps because the language learner is driven towards the analysis because of anti-homophony, anti-symmetry, the general analysis of phrasal verbal morphology, anti-locality, cartography and height of merger as parameter.

We can now give a positive answer to the question in (61): an analysis that assumes one and the same *-rare* in passive and psv-potential constructions can indeed also account for the presence of *-rare* in active potentials by reordering the relative merge order of the silent modal and passive *-rare*. We take

[56]The passive forms are also compatible with the present tense, and the potential forms are also compatible with the past tense, as in English passives and potentials. However, for ease of processing, they are listed here with past-tense and present-tense forms, respectively.

[57]*korob-e-ru* 'fall' is ok but needs some context because people do not usually fall intentionally.

[58]This is similar to ciCewa *-ik* (see Dubinsky & Simango (1996, 4.1)). With change of state verbs, *-ik* derives stative predicates, with a broader range of predicates it yields (psv-)potentials *bite-able* (but not statives **bitten)*

[59]See Ariji (2006), Ishii (2012), Ishizuka (2010, 2012), Sugioka (1984), Matsumoto (1996), Nishigauchi (1993), Wurmbrand (2012), Yashima (2008), Zushi (2008), and others.

this to provide strong support for Anti-homophony, decomposition, our general approach to phrasal verbal morphology, Antisymmetry (phrasal leftward movements feed further syntactic derivation), Anti-locality, and height of merge as a parameter.

Unification of these phenomena is simply impossible for any theory where passive voice *directly* affects the basic argument structure of a predicate. Under such an analysis, a single *-rare* that unifies the passive, psv-potentials and active potentials is a non-starter. The external argument must be projected in the high-psv potential, which means standard analyses which propose the external argument in the passive is not projected syntactically ('absorbed,' lexically unlinked, or semantically bound), will fail to identify the core properties of a single *-rare*.

10 Extending the analysis to other *-rare* constructions

So far, we have shown that the same *-rare* occurs in passives and psv- and active potentials. However, *-rare* also occurs in subject honorifics, middle/spontaneous constructions, as well as in long passives and long potentials. In the latter case, we are clearly dealing with the same *-rare* that can be merged above a series of functional categories (Cinque 2022), and so can the silent MOD (see (93) in appendix C).

The question, however, remains whether the variables we have identified can capture subject honorific and middle constructions with *-rare*.

10.1 Subject honorific constructions with *-rare*

The morpheme *-rare* also occurs in subject honorific constructions, as exemplified below:

(66) Sensei-ga {hasi-rare/korob-are} -ta.
teacher-NOM {run-RARE/fall-(r)are} -PAST
'The teacher (honorably) ran/fell.'

(67) Sensei-ga hon-o kaw-are-ta.
teacher-NOM book-ACC buy-(R)ARE-PAST
'The teacher (honorably) bought a book.'

Subject honorifics raise several questions: what exactly is *-rare* doing in this construction? Given Anti-homophony, this must be the same passive *-rare*. If so, why is honorific *-rare*, unlike passive *-rare*, not sensitive to verb class, and why do we find active alignment of the arguments of the main predicate?

These puzzles fall into place in the analysis sketched in Ishizuka (2012, pp.46-48). She argues *-rare* is in fact the regular passive voice, taking a *v*P as its complement. This *v*P, however, is headed by a silent transitive predicate HONOR, which selects SPEAKER as its external argument and a clausal 'AspP' as its internal argument. This leads to the representation in (68-a), and subsequent steps in the derivation.

(68) a. Build *v*P, and Merge *-rare* with *v*P ⇒
-rare [$_{vP}$ PRO$_{speaker}$ *v* [$_{VP}$ [$_{AspP}$ teacher *hasi*(=run)] HONOR]]
b. Form suffix:(Satisfy [epp[+V]] of *-rare*). IM VP, since *v*P is frozen.
[$\boxed{\text{VP[}_{\text{AspP}}}$ $\boxed{\text{teacher}}$ *hasi*(=run)] HONOR] [*-rare* [$_{vP}$ PRO$_{speaker}$ *v* [...]]]]
c. EM *-ga*, IM closest DP to *-ga*

In (68-b), the complement of little *v*, VP merges with *-rare*, satisfying its suffixal property, and yielding the linear order .... *hasi]* HONOR *] -rare*. This movement also 'demotes' the silent external argument SPEAKER in Spec,*v*P, as it always is in the immediate complement of *-rare*. In the step in (68-c) the embedded (honorific) subject raises to nominative *-ga*. We arrive in (68-c) at the starting point of the raising to subject analysis of subject honorifics, argued for by Hasegawa (1988), Kubo (1992), Toyoshima (2008). In sum, the selectional restrictions of *-rare* are satisfied by the silent transitive predicate HONOR, and the V in the AspP is not (and cannot be) subject to any restrictions imposed by *-rare*, because there is no merge relation between *-rare* and V. Finally, the resulting sentences have active alignment (as in (67)), because of the active alignment embedded in the small clause/AspP of HONOR, out of which a honorific subject has raised. We thus conclude that the same passive *-rare* also appears to occur in subject honorific constructions, though further details remain to be worked out.

10.2 Middle constructions

Next we turn to *-rare* constructions that occur *lower* than the volitional agent-theme passive. These are (various) middles, which have in common that *-rare* attaches to transitives or stems with a cause external argument. Though we did not investigate this thoroughly, oblique passives appear to be excluded from middles. While medio-passive middles are not problematic for our analysis (since these have a cause external argument, and a theme), so-called spontaneous middles are the most problematic cases for the extension of our syntactic approach:[60]

(69) Siiru-ga (*Ken-ni) hag -are -ta.
sticker-NOM (*Ken-DAT) peel -(R)ARE -PAST
'The sticker came off (by itself/*by Ken).'

It is generally assumed that the external argument is not syntactically projected for spontaneous middles, and that these start out syntactically as intransitive verbs. Yet, given the logic of our approach, and the incorporation of morphology in the syntax (there would be no way to derive an intransitive verb from a transitive verb), the complement of *-rare* in such cases must minimally count as *syntactically* transitive, i.e. have an external argument.[61] This raises the difficult question what the external argument in such cases could be. We tentatively suggest it is equal to "some (indefinite/unknown) force." It remains unclear, however, how to support this empirically. So, at this point this remains more like an hypothesis.

We have thus shown that it is possible to construct an invariant lexical item for passive Voice in passives, psv- and active potentials, long passives and potentials, subject honorifics, medio-passives, and hypothetically for spontaneous middles, where the different properties are due to differences in height of merge in a strongly decomposed clausal spine. The distributional differences in different constructions, therefore, are not due to the division of labor between different components, such as lexicon/pre-syntactic morphology vs. syntax or syntax vs. semantics, or syntax and post-syntactic morphology. In syntactic accounts of morphology, from DM,[62] to Nanosyntax,[63] and Morphology-as-Syntax (Collins & Kayne 2021, Koopman 2014), this is widely seen as following from differences in the size of the complement of passive, and variations in the height at which it may merge in the spine. This is the general account for the well-known

[60] We note that *-ar-* often appears in the middle with a reflexive meaning (*kurum-u* 'wrap,' *kurum-ar-u* 'wrap oneself'). This suggests further syntactic decomposition of *(r)ar-e*. *(r)ar-e* does not allow for a reflexive meaning.

[61] For (69), the stem is uncontroversially transitive, since the root *hag(-u)* 'to skin' serves as a transitive verb, which can combine with a middle (69), and with causative *-(s)as* and regular passive *-(r)are* (i.e, *hag-as-u* peel-CAUS-PRS 'take off,' *hag-as-are-ru* 'peel-CAUS-PASS-PRS 'be taken off').

[62] Halle & Marantz (1993), Harley & Noyer (1999), Embick & Noyer (2007), Bobaljik (2017).

[63] Caha (2009), Starke (2009), Baunaz et al. (2018, ch.1).

differences between adjectival and verbal passives (Embick 2004); between different types of causatives (lexical vs. syntactic) (Pylkkänen 2008, and others); between modals (cf. Cinque 2006, ch.7, for the root and alethic modal *yAbIl* in Turkish); between different passive-like Voices (Cinque 2022); between different *-able* constructions in English (Kayne 1984a), and beyond (Oltra-Massuet 2008, and others). Our account thus squarely falls within this general enterprise.

10.3 Generalized 'Smuggling': Tough Constructions in Japanese

So far, "smuggling" in passive and psv-potentials is the result of (i) the suffix *-rare* demanding a verbal phrasal constituent containing a verb as its Spec. This interacts with (ii) Anti-locality, which freezes *-rare*'s immediate complement; and (iii) Attract closest, which attracts the closest verbal projection, i.e. the complement of *-rare*'s immediate complement, and feeds further movements of caseless DP to *-ga*. This account predicts we should also find passive-like phenomena in environments that involve other phrasal affixes with similar complementation properties, not just passive voice. We outline such an account for bound adjectival predicates like *-yasu(i)* 'easy,' *-niku(i)* 'hard' in Japanese *tough* movement constructions.[64]

Kuroda (see 1992, 2012) explored similarities (and differences) between potentials and complex adjectivals. This was instrumental in establishing potentials as a type of tough construction. Furthermore, Chomsky's influential analysis of tough movement as involving a step of A' movement (Chomsky 1977, 1981) put the research focus on the (still problematic) tough movement analysis and turned the attention away from the parallels (and differences) of potentials and passives. Here we highlight the shared properties of passives and psv-potentials, in light of our analysis. Themes can map onto descriptive *-ga* (just as in theme psv-potentials), and, important from our perspective, in root clauses locative adjuncts and instrumentals can map onto the high (exhaustive) *-ga*, or in embedded environments on descriptive *-ga*, crossing over PRO, as shown below. In non-root clauses, the nominative locatives and instrumentals[65] do not get an exhaustive interpretation, again, similar to psv-potentials.

(70) a. Kono basyo-ga kuruma-o tome-yasu-i.
this place-NOM car-ACC park-easy-PRS
'This place is easy to PRO park the car at.' — Locative, Root-exhaustive *-ga*

b. Sono pen-ga zi -o kaki -niku -i.
that pen-NOM letter -ACC write -tough -PRS
'That pen is difficult to PRO write letters with.' — Instrumental, Root-exhaustive *-ga*

c. [Mosi kono basyo-ga kuruma-o tome-yasu-i -nara] . . .
[If this place-NOM- car-ACC park-easy-PRS -then] . . .
'If it is easy to PRO park the car at this place, . . . ' — Locative, non-root descriptive *-ga*

Given our analysis, these tough-movement properties fall out from (i) their bound nature (from predicate raising/complex predicate formation, as traditionally assumed in these constructions), and (ii) the fact that they select similar sized complement(s) as potential *-rare* does: *v*P or a clausal AspP, and the locations where different Obliques an merge. Anti-locality freezes their immediate complement, stranding the embedded

[64] See Inoue (2004, 1978), Saito (1982) Takezawa (1987), Kuroda (1992), and fn 65.

[65] Kuroda (1992, ch.8,p.270) acknowledges that his analysis for potentials (control: NP_i [NP_i . . . V] *eru*) (which only covers *ability potentials*) does not cover oblique psv-potentials with locative and instrumental nominatives.

subject (PRO). In case the A takes a *v*P complement, a phrase containing a verb (VP) I-merges to the spec of the adjective, smuggling the theme to descriptive *-ga*. In case it takes an AspP complement, the complement of the frozen projection I-merges to the Spec of the A, allowing the phrase containing IM of the highest DP/adjunct to high *-ga* for oblique nominatives (exhaustive only in root contexts (cf. Saito 1982, Kuroda, Ch.8 p.270). This thus revives a passive-like account for tough-constructions, though a full analysis of the four types of tough-constructions identified in Inoue (2004) along these lines remains to be investigated in the future. In §11.2, we return to another case of 'generalized smuggling' involving PPs, solving an apparent paradox that will become clear in the next section.

11 Comparative Syntax: Oblique DPs and PPs

Here we return to a central part of our proposal for Japanese: the external argument is introduced below instrumental and locative adjuncts and passive voice. We contrast our hierarchy with the different influential proposal made by Pylkkänen (2008) and turn to crosslinguistic support for our analysis from Malagasy (Austronesian) and ciCewa (Bantu). This has further consequences for the general understanding of the nature of Voice.

11.1 Crosslinguistic Support for *Loc> Instr> Act/Pass Voice>* Agent

A central feature of our account is that the basic Passive Voice in Japanese is merged *below* locatives and instrumentals (see (24)). Since we took canonical Passive Voice to merge with *v*P, with *v* introducing the agent, we also assume the external argument to be introduced *below* locatives and instrumentals in active sentences, as in (71-a). Our proposal differs from Pylkkänen's (2008) influential analysis, who, following Kratzer (1996), assumes that the external argument is introduced by a (silent) Active Voice head. Crucially, she locates Active Voice *above* what she calls "high" Applicatives (Appl), as in (71-c). The latter includes instrumentals, locatives and high benefactives in Bantu. Passive Voice, Pylkkänen proposes, is in complementary distribution with (silent) Active Voice (for volitional agents), but does not project a specifier. If this is universal, this leads to the expectation that Active and Passive Voice merge *above* locatives and instrumentals in Japanese as well, undermining our analysis. We agree with Pylkkänen that there must be a subject position *above* Loc> Instr (see (53)). This is because the subject maps onto the nominative in active sentences in the presence of locative and instrumental applicative/PPs, and the external argument or theme can bind into locative and instrumentals (see also Ngonyani 1996). It does not tell us though if this is a thematic position (as she assumes (71-c)) or a non-thematic position (as we assume (71-a)).

(71) a. Japanese: (active) $S_{[-\theta]}$ > Loc> Instr > (Act) Voice >[$EA_{[+\theta]}$ *v* VP]
b. Japanese: (passive) > Loc> Instr > $Pass_{rare}$ >[$EA_{[+\theta]}$ *v* VP]

c. Pylkkänen: (active) ... $EA_{[+\theta]}$ Act.Voice > Loc/Instr Appl> VP
d. Pylkkänen: (passive) Pass Voice > Loc/Instr Appl> VP

The issues are (i) where exactly the thematic position of the external argument is located, and for us (ii) how the movement of the external argument avoids a Minimality violation in (71-a), to which we return in §11.2.

We next present strong empirical evidence from the Voice system of Malagasy (Western Austronesian) and from ciCewa (Bantu) in support of a thematic position *below* the locative and instrumental. Specifically, Malagasy voices show that Active Voice is merged *below* instrumental and locative 'passives.' ciCewa shows that Passive Voice (hence Active Voice introducing the external argument) is merged prior to the formation of locative or instrumental applicatives.

11.1.1 Malagasy

Malagasy (Western Austronesian) has, like many Philippine languages, an elaborate Voice marking system, which is central to understanding its syntax. All Malagasy verbs (except some bare, non-active roots) are marked for **Active Voice (AV)** (also referred to as Actor[66] Voice or Actor Topic), **Theme Voice (TV)**, or oblique Voice or **Circumstantial Voice (CrcV)**, which promotes different oblique arguments or adjuncts to a predicate external position, (where it is uncontroversial that 'predicate' refers to a large (clausal-like) constituent (going back to Keenan & Li 1976, Keenan 1995, and many others). As in Japanese (and in Bantu), obliques can be expressed by DPs, which interact with the Voice system, or full PPs, which do not.

The predicate external position we are interested in here is the right peripheral 'S' position found in *VO....S* sentences, where S is marked with nominative Case, and can correspond to an agent, theme, goal, instrumental or locative, depending on the Voice morphology.[67] We concentrate on S, as opposed to the left peripheral focus position or the head of relatives, which appear to accept a broader class of obliques (including temporals, etc) using the same Voice morphology. The following two properties are relevant for our discussion: (i) the relation of CrcV with the AV and TV morphology, and (ii) the function of the arguments and obliques that map onto the right peripheral position depending on the different Voices.

Morphologically, the Voices are composed as follows (see Keenan, Ralalaoherivony & Fils Ranaivoson 2025, for recent discussion): The AV form is a **prefix**: *aN-* or *i-*. The selection of the AV form depends on (often idiosyncratic) properties of the root. If a root can occur with either *aN-* or *i-*, the *-aN* form tends to be transitive. We note that *i-* also seems to be the default form that occurs on loans. The TV form has several allomorphs: a **suffix** *-in(a)* or *-an(a)*, in some listed cases a prefix *a-* or a bare root. Finally, the CrcV is an invariant **suffix** *-an(a)* stacked on top of the AV form when used for instrumental and locative subjects.[68]

This is illustrated in (72) for the transitive forms of two roots: *vaki* 'read, open' and *vidi* 'buy.' These take different AV prefixes, the same TV suffix *-in(a)* (not *-an(a)*), but an invariant *-an(a)* suffix for the CrcV, stacked on the AV form.

(72)

root	Active Voice	Theme Voice	Circumstantial Voice(Loc/Instr)
/vaki/	m.a-maky	vaki-in(a)	[[a-maki] -an(a)]
/vidi/	m.i-vidy	vidi-ina	[[i-vidi] -an(a)]

[66]"Actor" in the sense of Schachter (1976) corresponds to the highest argument in the active. It is not necessarily an agent, but can be an agent, experiencer, unaccusative theme, etc, just as in active sentences in languages that do not overtly mark Active Voice. This distinguishes AV from the (unfortunately chosen label) (Active) Voice in the event semantics approach of Kratzer (1996), Pylkkänen (2008) and many others that use the label to indicate the node that introduces the (volitional) agent of a highly transitive verb.

[67]We treat these as "passives," but see Pearson (2005) who treats them as topics. The presence of the external argument is a shared feature with Pearson's (2005).

[68]See Keenan et al. (2025, ex (1)) for some verbs that distinguish a goal voice for goal subjects, and CrcV for instrumental and locatives.

The CrcV suffix is insensitive to the idiosyncratic lexical properties of the root and does not show *i+a=e* coalescence with *i* final roots, which applies on the root+$TV_{an(a)}$ level.[69] This points to the morphological structure in [[AV-root/stem] *-an(a)*]. The CrcV maps goal/benefactive, instrumental, or locative obliques DPs onto the right peripheral "subject/topic" position, as is well known from the literature. The morphology, therefore, shows that in order to have a locative or instrumental subject, an Active Voice form must first be constructed. The highest argument of the predicate is silent ('demoted') or preceded by a nasal linker−a clearly nominal property.

(73) ***-an(a)***$_{CrcV}$ . . . > Instr/Loc> **Active Voice** > *v*/V

We can thus conclude from the morphosyntactic structure that these oblique arguments/adjuncts are merged above Active Voice, just as we proposed for Japanese.

11.1.2 ciCewa: Locative and Instrumental applicatives and Passive

ciCewa has the extensive verbal morphology characteristic of Bantu languages, which figure prominently in Pylkkänen (2008). Here we focus on the (single) postverbal applicative morpheme *-ir-il* and the (single) passive morpheme *-idw*, which all harmonize (i/e) with the vowel of the stem. Most of the extensive literature[70] focuses on passives of benefactive applicatives, with sporadic mention of instrumental and locative passives. Our discussion below relies on the excellent work of Simango's (1995), in particular his in-depth systematic investigation of instrumental and locative applicatives, which had not previously been studied in similar detail.[71]

The applicative morpheme (Appl_{-ir}) introduces different oblique arguments or adjuncts. We will use $\textit{-ir}_{ben/instr/loc}$ or $\mathrm{Appl}_{ben/instr/loc}$ in order to identify the type of oblique argument/adjunct it introduces. Only one Appl can occur in the verbal complex.

(74) a. -ir_{ben} introduces benefactive DPs. The applicative is the only way to express a benefactive.
b. -ir_{inst} introduces instrumental DPs. Without the applicative the instrumental must be a full PPs.
c. -ir_{loc} introduces locative adjuncts (event modifiers), and must cooccur with locative DPs/PPs (which can also be present in the absence of the APPL.

The different interpretations of the Appl correlate with different syntactic properties, such as linear orders, primary object properties, morpheme orders (between Passive and Appl), promotability in passive constructions, VP ellipsis (Ngonyani 1996). We assume interpretations are determined by their syntactic environment: Appl_{-ir} can merge at different heights, yielding different interpretations according to a (probably) universal hierarchy (see also Japanese).

ciCewa has not only an applied suffix, but also a passive suffix. Passivization requires a transitive base (with a (volitional) agent and theme) Passivization of (non applicativized) unergatives is disallowed

[69] Note that some Malagasy roots have a final floating Cs. These will associate in the CrcV form in case of any vowel initial verbal suffix. This is clearly a late phonological rule.

[70] See Alsina & Mchombo (1990), Alsina (1999), Dubinsky & Simango (1996), Simango (1995), Hyman (2003), Bresnan & Kanerva (1989) and many others.

[71] Japanese appears not to have high *ni*-benefactives, *no-tame-ni* GEN-BENEFIT-DAT 'for the sake of' is used instead.

(Simango 1995). Passive Voice thus shows the location of the external argument.[72] Relevant for our discussion here are the relative orders of the Pass and Appl suffixes and what can passivize: with the benefactive, only the V-Appl$_{ben}$-Pass order is allowed, and only the benefactive can passivize. This means the benefactive applicative must be formed before passivization applies. The locative and instrumental Appls show a different order with passive. For instrumentals, the V-Appl$_{inst}$-Pass is only allowed for unergative verbs, with promotion of the instrumental DP. With transitive verbs only the order V-pass-Appl$_{inst}$ is possible,[73] and the instrumental cannot be promoted. For locative, only the order V-pass-Appl$_{loc}$ is allowed. This will be discussed below. In anticipation, we summarize our proposal for the Pass/Appl hierarchy in ciCewa:

(75) ciCewa hierarchy: Appl$_{loc}$ > Pass $_{idw}$ > Appl$_{inst}$ > Pass $_{idw}$ > v > Appl$_{ben}$ [VP]

This hierarchy is consistent with the locative and instrumental obliques we proposed for Japanese (Loc > Instr > . . . ^{L}Pass$_{rare}$> EA: see (18) and (71-a)), and thus brings independent support for our analysis.

We now turn to a brief discussion of the interaction of passives with instrumental and locative applicatves. Instrumental passives are only possible with unergative verbs, in which case the V-Appl$_{instr}$-Pass$_{idw}$ order is required, as in (76):

(76) ✓ **Instrumental passive of 'unergative'** (Simango 1995, p.83)

Ndodo i- na- yend -er -edw -a (ndi mnyamata)
4.stick 4SM- PST- walk -APPL -PASS -FV (by 1.boy)
'The stick was walked with (by the boy).' $\boxed{\text{V-Appl-Pass}}$

Importantly, the V-Pass$_{idw}$-Appl$_{instr}$ order is also possible, but can only lead to theme passives, as in (77):

(77) **Theme passive** only (Simango 1995, p.45)

Anyamata a- na- kwapul -idw -ir -a chikoti (ndi Joza)
boys SB- PST- whip -PASS -APPL -FV cane (by Joza).
'The boys were whipped with a cane (by Joza).' $\boxed{\text{V-Pass-Appl}}$

The suffix order shows passivization takes place before an instrumental applicative is built.

As for locative passives, Simango argues that there are no locative passives in ciCewa, even though (78) looks like one: a locative DP can occupy the subject position, and both Pass and the Appl$_{loc}$ are present.

(78) Mu.nyumba mu-na- phik -idw -ir -a nyemba (ndi Joyce)
18.house 18S-PST- cook -PASS -APPL -FV 10.beans (by Joyce)
'The house was cooked beans (in) (by Joyce).' $\boxed{\text{V-Pass-Appl}_{loc}}$

The morpheme order corresponds to V-Pass$_{idw}$-Appl$_{loc}$, with Pass as the inner affix, and Appl$_{loc}$ as the outer one. Simango convincingly argues this sentence is not derived by passivization of the locative. Rather, there are two different processes involved in its derivation. As the morphology indicates, the passive first merges with vP (this accounts for the fact the verb *must* be transitive). Next Appl$_{loc}$ adds a locative adjunct DP

[72] Pylkkänen (2008) assumes Passive Voice and Active Voice are occupying the same position, and builds the agent requirement into the semantics (see Collins 2024, for critical discussion). Bruening (2013) assumes Passive Voice merges with a non-argument introducing Active Voice, (our) v.

[73] Simango (1995, 45,fn 6) addresses examples from the literature with V-Appl$_{inst}$/Appl$_{loc}$-Pass order, and states these sound odd for him and others regardless of what is promoted.

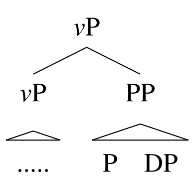
on the passivized verb. Finally, locative inversion (a case of predicate inversion), ubiquitous in Bantu, and known to apply only to unaccusative and passivized verbs in ciCewa (see Bresnan & Kanerva 1989, among others), moves the locative DP to the subject position, where it triggers subject agreement.

In conclusion, we see that ciCewa high applicatives do not form a single class (contra Pylkkänen 2008). While the benefactive applicative is always formed below the passive and the external argument, the instrumental and locative applicatives are formed above the passive, (hence above external argument), as shown by the constraints on passivization. The latter independently support our proposal that instrumental and locative adjuncts in Japanese cannot be passivized because passivization precedes the formation of locatives and instrumentals (i.e, Loc>Instr>Pass). See §11.3 for further discussion regarding the predictions of our proposal for a crosslinguistic typology of oblique passives (i.e. pseudopassives).

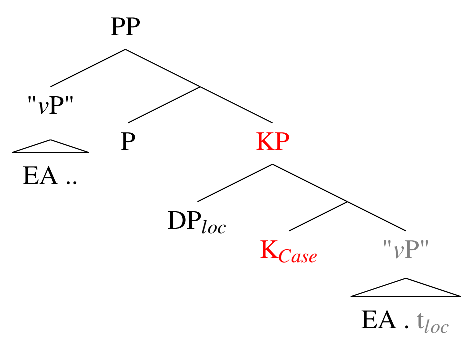
11.2 On Oblique DPs and Full PPs

We next address how it is possible for locatives and instrumentals to start out *above* the external argument, but to end up c-commanded by the nominative or the accusative, when they are full PPs (71-a). We can make sense of this by adopting Kayne's proposal for how PPs are built. Kayne (2000, chs.12-15; 2005, chs. 5,7,9) argues that CPs and PPs headed by a functional P or C are not base-generated constituents, as standardly assumed in traditional grammar and phrase structure grammars. Rather, they are the output of the syntax, built from two separate pieces brought together by I-merge, and 'look like' constituents at the end of the derivation. (79) shows both the traditional structure for a PP locative adjunct (79-a), which Kayne rejects, and his proposal (79-b). Importantly, the general schema in (79-b) applies to CP as well.

(79) a. Not the input to the syntax:



b. ...but the output:



In a derivation like (79-b), K (=Case) merges with the "*v*P," and attracts the closest DP. P merges with KP, and the complement of K, the remnant "*v*P", merges with P. "*v*P," P and K together determine the Case on K. The derivation in (79-b) is consistent with anti-locality: KP cannot move, but the complement of K, i.e. *v*P, can. *v*P movement to Spec, P thus "smuggles" the external argument in the *v*P over the oblique DP$_{loc}$. Though the derived surface constituencies in (79) may look indistinguishable, they make different empirical predictions. Kayne (op. cited) presents strong empirical support in favor of derivations like (79-b), as it allows relating the height at which Ps can merge with a variety of linear strings and different constructions.[74]

[74]Kayne's empirical support for the derivation includes (i) an account for the surprising subject object asymmetries found in Italian infinitival clauses headed by a prepositional C (see also Angelopoulos 2019, for Greek); (ii) a much better empirical account for positions in which Ps can be stranded, and a way of capturing the intra- and inter-variability of English speakers found in this respect. And last but not least, Kayne argues the same mechanism successfully derives a variety of apparent rightward movements, including (iii) extraposition of PPs from DP, (iv) extraposition of PPs, (v) PP scrambling, and (vi) extraposition of CPs and relative

The derivation in (79-b) also resolves the apparent conflict that arises in our analysis. We have proposed in (71-a) that locative or instrumental DPs are introduced *above* the external argument in the *v*P, but the external argument (and the accusative theme) ends up c-commanding the oblique PP in the active sentence. However, this is exactly what (79-b) shows: the oblique DP is merged above the *v*P and attracted to K (..DP_{obl}..). When the P layer is merged, the (verbal) complement of K raises to Spec,PP. Since *v*P is closest to the nominative subject position, the external argument maps onto the subject position, and ends up c-commanding the PP. This derivation thus not only derives the surface syntactic structure: it gets around the minimality problem caused by the oblique DP, and effectively "demotes" postverbal PPs. In this sense, oblique PPs (correctly) end up being the lowest "chômeurs" in relational grammar terms, with predicate movements 'feeding' further movements, and full PPs not interacting with the nominative/accusative case systems.

11.3 Towards a typology of oblique passives

A final point we make here is that our analysis bears on further questions as to where passive Voice can be merged in a given language, how this is constrained, and whether there are implicational hierarchies.

Using the universal hierarchy of the obliques in (80), and variation in height of merge for the passive, we summarize our findings in Japanese, Malagasy (non-active voice) and ciCewa oblique passives in (80).[75]

(80) Oblique Hierarchy: Locative > Instrumental > (Benefactive) > Goal > (Directional / Source)

a. Japanese: $\text{MOD}_{possibility}$ $\boxed{\text{Pass}}$ Loc Instr ($\text{MOD}_{ability}$) $\boxed{\text{Pass}}$ v Goal Dir/Source[76]
b. Malagasy: $\boxed{\text{CrcV}}$ Loc Instr $\boxed{\text{"Pass"}}$[77] Ben Goal $\boxed{\text{Pass}_{TV}\text{/Act}}$ v[78]
c. ciCewa: Loc $\boxed{\text{Pass}}$ Instr $\boxed{\text{Pass}}$ Ben v Goal[79]

In (81), we generate a preliminary oblique passive typology, as part of a more general typology of passive(-like) constructions, consisting of implicational hierarchies in (81) (a-d), and potential merge sites for passive Voice indicated with * in (81-e).

(81) a. If a language allows Goal passives, then it allows Theme passive (i.e. canonical passives).
b. If a language allows Benefactive passives, then it allows Goal passives.
c. If a language allows Instrumental passives, then it allows Benefactive and Goal passives
d. If a language allows Locative (adjunct) passives, then it allows all lower oblique passives, etc.
e. * Loc * Instr * (Ben) * Goal * (Directional/source)

This hierarchy makes clear testable predictions, which will have to be verified against a future database that details the crosslinguistic variation that we observe in this respect. (81-a) represents a language like English with goal passives, versus Dutch which lacks goal passives (except for very few listed predicates),

clauses.

[75]Possessor raising passives, which are ubiquitous in Japanese, Malagasy, and ciCewa, will have to be integrated. We leave out directional/source obliques in Malagasy, for which we have incomplete data.

[76]Recall that Japanese appears not to have high benefective DP obliques: a full PP *no-tame-ni* GEN-BENEFIT-DAT 'for the sake of' is used instead (see fn. 71).

[77]Some verbs have a distinct non-active form for ben/goal passives, many use the CrcV.

[78]It is unclear if TV can ever combine with CrcV.

[79]Appl_{loc} do not combine with directional oblique PPs (Simango 2012) unless the verb is unaccusative. Directionals combining with 'movement verbs' undergo locative inversion (Bresnan & Kanerva 1989).

and only allows theme passives. (81-b) represents a language which would allow passives of goals, but not of benefectives. This might represent some English varieties, ie. English speakers who accept benefactive passives versus those who do not. This should not only account for the kind of oblique passives a language allows, but also identify languages that lack oblique passives/pseudopassives altogether.

12 Conclusion and Further Perspectives

This paper investigated whether the same *-rare* occurs in the passive and psv-potential, as expected if there is a UG principle of Anti-homophony. We have shown that the same passive voice *-rare* occurs in the two constructions, and that (psv-)potentials in addition contain a silent modal MOD. This allowed for a systematic comparison of the forms, interpretations and syntactic effects of *-rare* in the two constructions. Differences should fall out in a modular fashion from our underlying theoretical assumptions, with Anti-symmetry playing in important role in simplifying and restricting possible analyses, and leftward phrasal movement feeding I-merge, and deriving morpheme orders.[80] and the interaction of *-rare* with other elements in its local environment. From the detailed structural map thus obtained, we should be able to determine the properties of one *-rare*, thus testing both the validity of the Principle of Anti-homophony, and the power of the theoretical assumptions that drive the analysis.

Differences in form: The form of *-rare* in the potential is sensitive to whether the stem ends in a consonant. *-rare* is invariant elsewhere. A late spell-out rule under strict locality of MOD captures this distribution.

Two flavors of potential modality: We have shown MOD comes in two flavors: (roughly) $\text{MOD}_{ability}$ and $\text{MOD}_{possibility}$. We established these are merged in different regions of the clause, bringing further support for the way they are incorporated in the hierarchy in Cinque (1999, 2006:ch.7).

Syntactic differences. We established that the type of oblique passives possible in the passive, and the psv-$\text{MOD}_{ability}$-potentials (goal, directional, source) belong to the big VP domain. Psv-$\text{MOD}_{possibility}$ potentials allow locative and instrumental passives in addition. These are *excluded* in regular passives and psv-$\text{MOD}_{ability}$-potentials.[81]

A substantial part of the paper was devoted to establishing the local environments and hierarchies involving passive and pvs-potential *-rare* so as to be able to extract the properties of one *-rare*. For the basic passives, we determined the height at which it can be minimally merged as (21), repeated below:

(82) Minimal height of merge for passive voice:
Pass_{rare} [$_{v\text{P}}$ ext.arg [$v_{cause/do}$ [$_{\text{“big”VP}}$ V .....]]]

Goal, Dir and Source obliques occur in the big VP domain (which remains to be further investigated) while locative, and instrumental obliques, we argued, must be merged *above* the basic passive voice, and the external argument (contra Pylkkänen (2008).[82] Further incorporating the psv-potential modalities (Mod

[80] Probably the biggest differences (beyond notation and Antisymmetry) with narrow syntax Minimalism are rich decomposed syntactic structures, strict locality of selection under Merge, and the incorporation of morphology in syntax, relying on leftwards phrasal movement.

[81] We further described (but did not analyze) the (im)possible realization of the silent external argument as a *ni*-(by)-phrase, (sometimes dependent on its content). Though further analytical work is required, we did establish that a *ni*-phrase is excluded in psv-$\text{MOD}_{possibility}$ potentials, and that the silent DP must get a generic interpretation, as impersonal passives cross-linguistically do.

[82] This merge order is further supported by Japanese internal evidence as well as by crosslinguistic evidence from Malagasy and ciCewa (see §11.1).

>Pass), and the linear orders of perf/progr *-tei* and *-rare* (cf.§6) brings us to (83):

(83) Psv-MOD_{poss} and Psv-MOD_{abl} potentials, oblique DPs, and linear orders of Perf/Progr *tei* and *-rare*:
..$\boxed{\text{MOD}_{poss} > \text{Pass}_{rare}}$ > ...Perf/Prog_{tei}...**Loc>Instr**...> $\boxed{\text{MOD}_{abl} > \text{Pass}_{rare}}$ [$_{vP}$ agent..[..**Goal>Dir/Src**]

(83) accounts for why low obliques can in principle be passivized, but locative and instrumental cannot. Locatives and instrumental passives are only possible if they are c-commanded by the high *-rare* that is part of high *possibility* psv-potentials. Further confirmation comes from the two possible orders of Perf/Progr *-tei* and *-rare*, which correlate with the type of oblique passives (see §6). Locative and instrumental (high) psv-potentials require the order *V-tei-rare*, passive and low *ability* psv-potentials require the order *V-rare-tei*. Finally, we have shown that the complement of psv-MOD_{poss} contains a silent generic PRO in a non-thematic position. This makes it comparable in size to certain infinitival complements in English. Given Cinque's hierarchy and the generic reading, we labeled this complement as HabP, as in (84-a). It differs in size from the complement of the basic passive or the low *ability* psv-potential (84-b). Since the Pass_{rare} in (84-a) does not interact with the argument structure of the predicate, an analysis where the passive directly affects the argument structure of a verb, cannot be extended to (84-b). Any analysis that takes the external argument to not be represented in the syntax therefore excludes a unified analysis for *-rare*. Extending (84-a) to (84-b) on the other hand makes them parallel, and allows for an analysis with one *-rare*.

(84) a. ..$\boxed{\text{MOD}_{poss} > \text{Pass}_{rare}}$ [$_{HabP}$ PRO_{gen} ...perf/prog_{tei}...**Loc>Instr** [....$_{vP/VP}$ t ....]]
b. ..$\boxed{(\text{MOD}_{abl} >)\ \text{Pass}_{rare}}$ [$_{vP}$ PRO_{agent}..[..**Goal>Dir/Source**]

We thus concluded the basic passive must have a syntactically projected silent agent−PRO−as well, and that its silence should be accounted for in the same way as PRO in certain infinitivals.

Analysis of passive and psv-potentials: We thus argued the same *-rare* occurs in passives and psv-potentials, and showed how it derives the different passives. As a phrasal affix, *-rare* must build a verbal specifier. Given Anti-locality (Abels 2003, Kayne 2005), which is widely adopted in Kayne's work (see §11.2,§10.3), its complement is frozen. The closest accessible phrase containing a verb is the complement of the frozen head. I-Merge to *-rare* thus satisfies the property of *-rare*. 'Smuggling' carries up the highest caseless DP in the domain of the nominative, feeding I-merge to *-ga*. The external argument is frozen under *-rare*, and has the distribution of PRO. It does not intervene, and is in effect demoted. No extra devices, lexical or other, are necessary to account for its presence, nor intervention or demotion.

Double/multiple *-ga*: The structure in (83) interacts with double/multiple *-ga* in Japanese (Kuno 1973, Kuroda 2012, a.o). We adopted (at least) two merge sites for *-ga*: a high, T dependent *-ga* (which gets an exhaustive reading in root-, but not in embedded environments), and a much lower *-ga* (which is not exhaustive even in root environments), which (only) theme passives oblique passives and theme potentials can map on. Only one *-ga* can surface on a given DP. [83] Consistent with our analysis as Case-driven A-movement to *-ga*, is the fact that it behaves as A-movement for reconstruction, and scope (§8.2). We showed how this analysis extends to **active potentials** by a simple reordering of Pass>Mod, paving the way

[83] Low obliques passives are distinct from theme passives, in that they can only map on the high nominative case, as we concluded from the obligatory exhaustive reading of low obliques in passive and potentials in root (but not in non-root) environments. We leave further details of the analysis for future research.

to a broader understanding of deponent verbs and ergativity. We discussed how our analysis extends to other *-rare* constructions, like subject honorific *-rare* (which following Ishizuka 2012), passivizes a silent grammaticalized transitive verb), as well as middle constructions, which are 'smaller' transitives than (vol) agent-theme passives.

Further crosslinguistic support for the Loc>Inst> Voice> Agent merge order comes from oblique nominatives in Malagasy Voice system, and the interaction of passive with instrumental and locative applicatives in ciCewa (see §11.1). Lastly, we discussed in §11.2 (i) how Kayne's analysis of PPs can explain why oblique DPs (crosslinguistically) interact with the Voice system, but corresponding full PPs do not. Kayne's analysis for PPs given in §11.2 in fact 'demotes' PPs, and accounts for why external arguments start out below obliques, but end up c-commanding full PPs. We also constructed a putative typology of oblique passives, and sketched testable predictions (see §11.3).

Our analysis was guided by very general and well-motivated theoretical assumptions, as well as the factors that have been identified as accounting for differences in constructions within a language, like the same item being merged at different structural heights. We put some pieces of the overall puzzle together, concentrating on motivating and establishing the overall cartography and issues like where exactly the external argument is introduced in the structure, what oblique passives are possible and why, and what this can tell us about the nature of passive Voice. However, many of the topics we discussed need further study, and many areas that are part of the sprawling passive literature remain to be integrated into the syntactic representations we have motivated so far. This is expected given the highly modular nature of passive constructions, and our focus on motivating the analysis of a single *rare*. Specifically, we did not address the analysis of *ni*-(by)phrases, though we hinted at possible hypotheses in various footnotes (see especially fn 4, and fn 36). Nor did we delve into the interpretive properties of the silent external argument,[84] beyond the general distinction between existential and generic/human readings, which clearly seems to be related to structural height crosslinguistically. While we have outlined how to incorporate exhaustive and descriptive *-ga* into the general hierarchy, we did not discuss how the overall Japanese Case system, or 'scrambling' is to be integrated. (but see Kim 2023) for an attempt. Finally, we did not enter into the semantics of 'possibility' and 'ability': ideally these readings should be derived by merging the same denotation at different structural heights. Nor did we examine the readings of perfect/progressive *-tei* (see Takamine 2010). These, like many further topics, should be incorporated into our general findings in future research.

[84]See Angelopoulos, Collins & Terzi (2020), Collins (2024) for a further list of diagnostics and references.

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Appendices

A The suppletive potential: *deki-ru*

The discussion has been restricted to potentials with the suffix *-(rar)e*. However, a special suppletive verb *deki-(ru)* occurs in the 'noun plus light verb' *su-(ru)* construction (not **si-rare*) in all the types of *-rare* potentials, i.e. in low and high psv-potentials and active potentials (see also Kanno 1992, Takano 2003).

(85) Theme passives

a. Sono mondai-ga (Mei-ni) kaiketu -deki -ru.
that problem-NOM Mei-DAT solution -DO.CAN -PRS
'That problem is solvable (by Mei).' ≈ That problem can be solved (by Mei). Low psv-poten

b. Sono e -ga (*?Ken-ni) kono bizyutukan-de kansyo -deki -ru.
that picture -NOM (*?Ken-DAT) this art.gallery-LOC see(appreciation) -DO.CAN -PRS
'That picture can be seen at this art gallery.' High psv-poten

(86) a. Kono bizyutukan-ga sono e -o kansyo -deki -ru.
this art.gallery-NOM that picture -ACC see(appreciation) -DO.CAN -PRS
'This art gallery can see that picture.' Locative psv-poten
'It is possible to see/appreciate that picture in this art gallery'

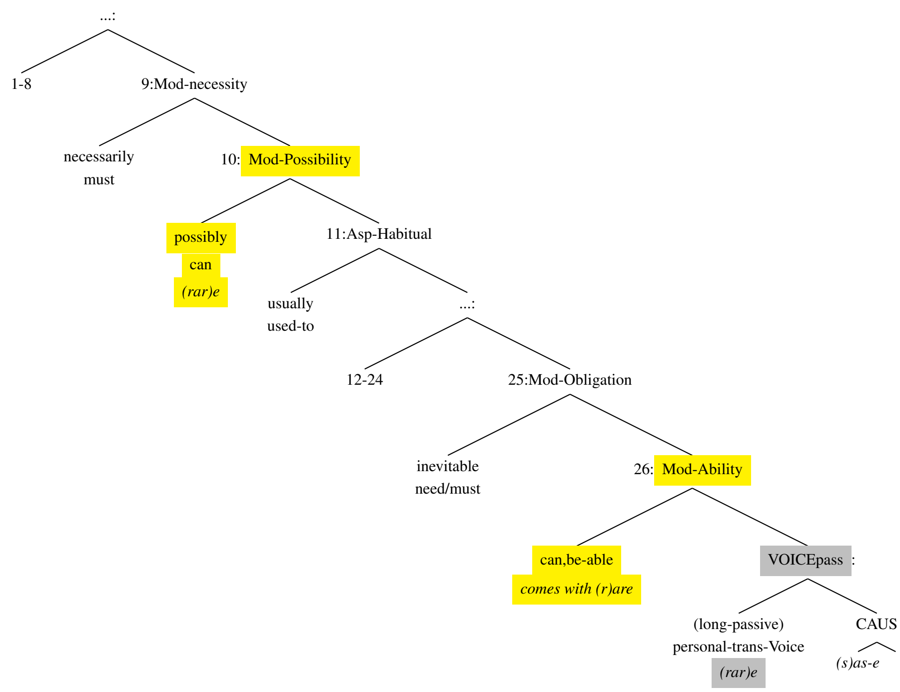
b. Naomi-ga sono mondai-o kaiketu -deki -ru.
Naomi-NOM that problem-ACC solution -DO.CAN -PRS
'Naomi can solve that problem.' Active Poten

The fact that the same *deki-(ru)* form is used in all types of potential constructions is strong evidence for the unified analysis of active and psv-potentials.

B A finer hierarchy

The tree in (87) is a finer hierarchy presented by Sportiche, Koopman & Stabler (2013, pp.369-370), which shows the left and right environments of the MOD. Numbers refer to the nodes in the tree. The Japanese potential does not appear to cover the Mod-necessity reading, neither the high one (9) nor the low one (25).

(87)



A considerable number of functional elements can intervene between the two MODs, shown above as 12-24 with corresponding light verbs or adverbial modifiers in English. We list the longer list 10-29 in (88), as it includes Mod-Permission.

(88) 10: Mod-Possibility$_{possibly,can}$ > 11: Asp-Habitual$_{usually,used.to}$ > 12: Asp-Predispositional$_{tend}$ > 13: Asp-Frequentative$_{often}$ > 14: Mod-Volitional$_{want,willingly}$ > 15: Asp-Celerative$_{quickly}$ > 16: Tense-Anterior$_{already}$ > 17: Asp-Terminative$_{any,no.longer}$ > 18: Asp-Continuative$_{still}$ > 19: Asp-Perfect$_{always}$ > 20: Asp-Retrospective$_{just}$ > 21: Asp-Proximate$_{soon}$ > 22: Asp-Durative$_{briefly,long,for/in.an.hour}$ > 23: Asp-Generic Progressive$_{characteristically,progressively}$ > 24: Asp-Prospective$_{almost,be.about.to}$ > 25: Mod-Obligation$_{inevitably,need,must}$ > 26: Mod-Ability$_{cleverly,can,be.able}$ > 27: Asp-Inceptive$_{begin}$ > 28: Asp-Frustrative/Success$_{manage}$ > . . . 29: Mod-Permission$_{be.allowed}$ > . . .

Japanese potentials with *-rare* can have a "permission" reading in addition to the possibility or ability reading, as pointed out in fn.22. Though in Cinque's hierarchy, 'permission' is below 'ability,' these readings in Japanese show the characteristic properties of high psv-potentials: they are possible with locative psv-potentials and do not allow a *ni* by-phrase, as illustrated below:

(89) Kono dooro-ga zisoku nanazyuk.kiro -de hasir -e -ru.
This road-NOM hourly 70.kilo -DE run -(RAR)E -PRS
'This road can be driven 70 km per hour on.' ≈ It is possible to drive 70 km per hour on this road.

This suggests the permission reading in such cases might be closer to *(legally) possible/"authorized,"* explaining the high potential properties (i.e, locative passives and impossibility of a *ni* by-phrase).

C Long passives and long potentials

Similar to long passives like (90), long potentials are also possible in Japanese (for long passives in Japanese see Ariji 2006, Fukuda 2012, Ishii 2012, Ishizuka 2010, 2012, Sugioka 1984, Matsumoto 1996, Nishigauchi 1993, Wurmbrand 2012, Yashima 2008, Zushi 2008, and for other languages, see Cinque 2004, 2006, 2021, and the long list of references cited therefin).

(90) **Long passive**

Kono hon-ga kodomo.tati -ni yomi-tuzuke **-rare** -tei -ru.
this book-NOM child.PL -DAT read-continue -RARE -PERF/PROGR -PRS
Lit. 'This book is continued to read by children.' $\boxed{\text{Theme}}$

(91) **Long active potential**

Ken-ga ringo-o tabe-tuzuke -rare -ru.
Ken-NOM apple-ACC eat-continue -RARE -PRS
'Ken is able to continue eating apples.' $\boxed{\text{Agent}}$

(92) **Long psv-potential**

a. **Low psv-potential**

Kono hon-ga go.sai.zi-ni yomi-tuzuke -rare -(?tei) -ru.
this book-NOM 5.year.old.child-DAT read-continue -RARE -PERF/PROGR -PRS
Lit. 'This book can be continued to read by a 5-year old.' $\boxed{\text{Theme}}$

b. **High psv-potential**

Kono karaoke.ya-ga (nanzikan-mo) uta-o utai-tuzuke -rare -ru.
this karaoke.shop-NOM (for.many.hours-also) song-ACC sing-continue -RARE -PRS
Lit.'This karaoke shop is possible to continue singing songs in (for many hours).' $\boxed{\text{Locative}}$

Here is an update of Ishizuka's (2012:p.37) list of restructuring verbs in Japanese, incorporating high & low MOD and Passive Voice into the universal hierarchy of functional categories proposed by Cinque (1999, 2004) (see also fn 34).[85]

(93) $\boxed{\text{A subset of the clausal hierarchy in Japanese}}$

$\ldots \underset{ta/ru}{\mathrm{T}_{past/prs}} > (\mathrm{Neg}_{na.\{i/kar\}}) > \ldots\ \underset{(rar)e}{\mathrm{MOD}_{possibility} > {}^{H}\mathbf{Voice}_{passive}} > \mathrm{Asp}_{habitual} > \underset{tuzuke}{\mathrm{Asp}_{continuative(I)}}$

$> (\underset{ta.\{i/gar\}}{\mathrm{Mod}_{volitional}}) > \underset{tei}{\mathrm{Asp}_{perfect}} > \underset{tei}{\mathrm{Asp}_{progress}} > \underset{kake}{\mathrm{Asp}_{prospective}} > (\underset{ta.\{i/gar\}}{\mathrm{Mod}_{volitional}}) > \underset{hazime,das}{\mathrm{Asp}_{Inceptive(I)}} >$

$\underset{(rar)e}{\mathrm{MOD}_{ability} > {}^{L}\mathbf{Voice}_{passive}} > \underset{sokone}{\mathrm{Asp}_{frustrative,success}} > \underset{oe,age,wasure}{\mathrm{Asp}_{completive(I)}} > \underset{(r)are}{\mathrm{Voice}_{passive}} > (\mathrm{Causative}_{(s)ase}) >$

$\ldots > \underset{hazime}{\mathrm{Asp}_{inceptive(II)}} > \underset{tuzuke}{\mathrm{Asp}_{continuative(II)}} > \underset{oe,wasure}{\mathrm{Asp}_{completive(II)}} > \underset{naos}{\mathrm{Asp}_{repetitive}} \ldots > \mathrm{V}$

Note that Japanese negation *na.i* and Mod$_{volition}$ *ta.i/gar* are adjectives. The latter do not necessarily match the positions proposed in Cinque's universal hierarchy. However, we include them in (93) for reference.

[85] Cinque (2004, 1999) has presented a couple of partial hierarchies, and this Japanese hierarchy is based on the universal hierarchy created by combining the four partial hierarchies given in Cinque (2004:p.24(3), p.76(16), p.93(64)) and Cinque (1999:p.81(12)).

These adjectives need to take verbalizers *kar/gar* to combine with, for example, the past tense suffix *-ta*. Further, Japanese Causative suffix *-(s)ase* appears not to be a pure grammatical functional head, since it can be iterated (see Cinque 2004: p.91, fn.18). Nevertheless, we include it in (93) since its relative order with respect to passive Voice *-rare* is more or less clear (see Ishizuka in prep for evidence and more discussion).

D Derivations showing "compression" or "bundling"

Here we address a technical problem with respect to the exact derived structure at spell out. As we saw, the spell out of passive *rare* is sensitive to whether a stem ends in C in the potential, and raises the question how this is read off our structures.

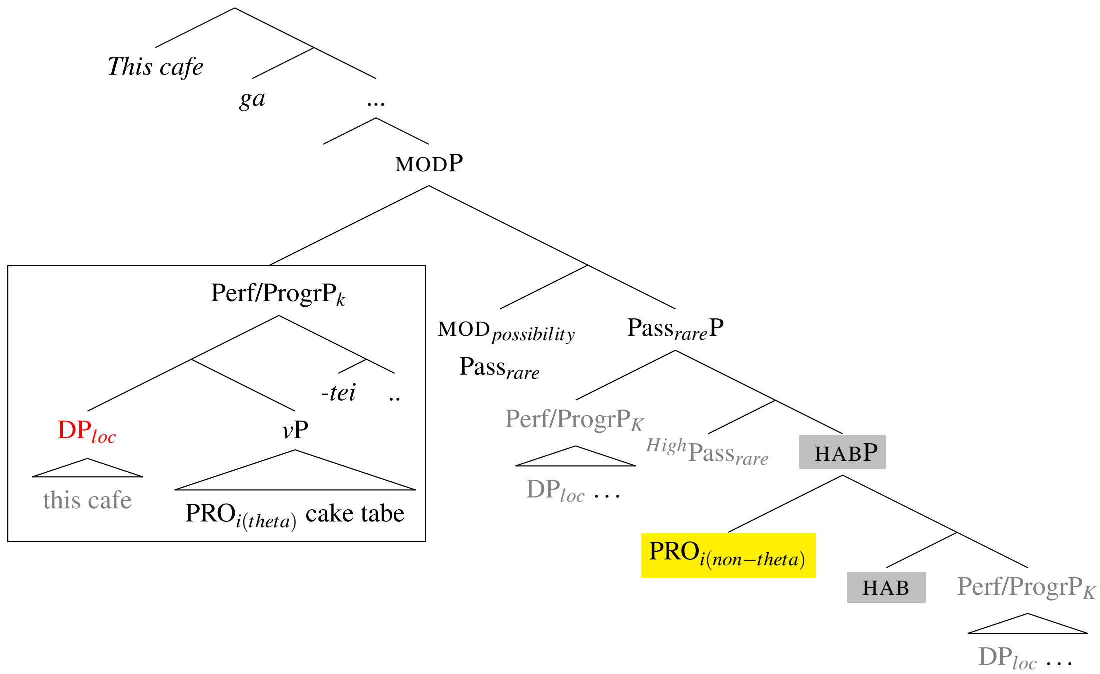
This problem arises in part because of the one-feature-per-projection hypothesis that we adopt, which yields the order of operations, but is not well equipped to express the cooccurence of specific projections, like the local dependency of the silent modal and *-rare*. We can solve this problem by assuming that featural complexes can be created by syntactic movement: they are the **output** of the syntactic derivation, not the **input** (see Koopman 2003, 2006). Concretely, we achieve this by "compressing" the two projections into one, by (i) a step of head movement of *-rare* to the silent MOD−the one context in which Koopman & Szabolcsi (2000, pp.4-5) allow head movement−and (ii) a step of Spec extraction of the VP_k, which merges to the left of the $\text{MOD}_{ability}$ and *-rare* complex, a step compatible with anti-locality.

D.1 High locative psv-potential

Incorporating the "Compression/Bundling" of MOD and *rare* for spell out reasons:

(94) head move *-rare* to $\text{MOD}_{possibility}$, I-merge Perf/ProgrP to Spec, $\text{MOD}_{possibility}$, merge *-ga*, I-merge DP_{loc} with *-ga* (i.e. Attract Closest):

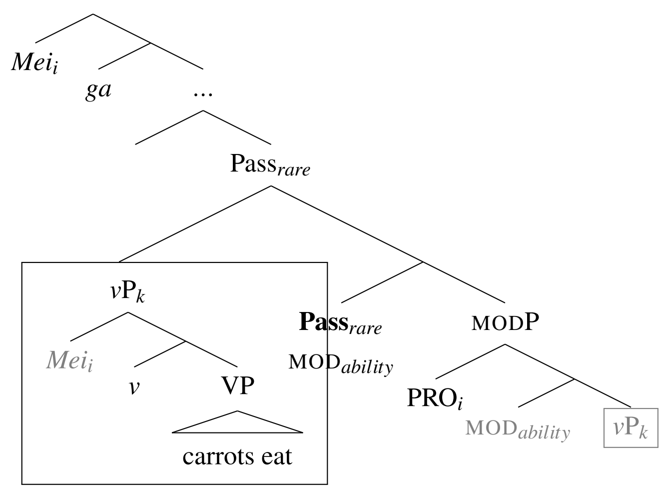
Kono kafe-ga keeki-o tabe -tei -rare (-ru)
this cafe-NOM cake-ACC eat -PERF/PROGR -RARE (-PRS)
Lit. 'It is possible to continue eating a cake in this cafe.'(One can stay for long time in this cafe eating just a single cake.)



D.2 A derivation of an active potential

The derivation of an active potential example is given in (95), starting from a Pass$_{rare}$>MOD merge order, with head movement of silent MOD to Pass$_{rare}$ ensuring the correct spell out for potential form.

(95) Mei-ga ninzin -o tabe-rare-ru.
M-NOM carrot -ACC eat-RARE-PRS
'Mei is able to eat carrots.'

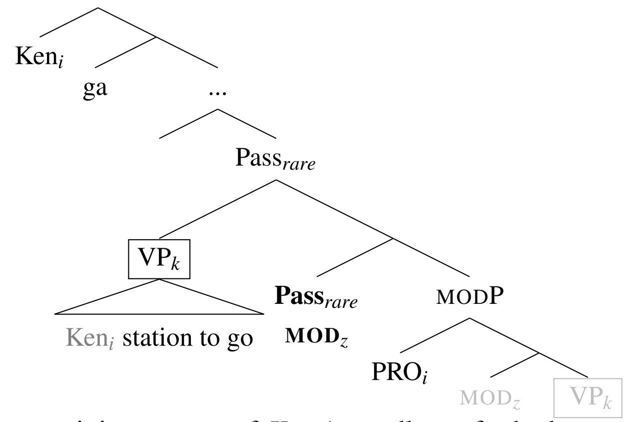


In (95), the complement property of *-rare* is satisfied by MOD$_{ability}$P which is frozen because of antilocality. The bound property of rare (i.e. its epp[+V] feature) is satisfied by I merge of *v*P, stranding MOD$_{ability}$P. The closest DP, *Mei*, raises to the nominative *ga*-position. This derivation therefore yields an active sentence with a passive morpheme in (62-a), performing its usual function.

D.3 A raising/control derivation for an unaccusative active potential

The derivation of (65-b) is provided below:

(96) Ken-ga eki-ni ik -(ar)e -ta.
Ken-NOM station-DAT go -(R)(AR)E -PAST
'Ken could go to the station.'



The derivation involves the step of VP movement, containing a copy of *Ken* (regardless of whether MOD is a Control or raising predicate), which will be further extracted to Spec, *-ga*. Given the conception of I-Merge as 'copy and delete under c-command,' the final structure looks unproblematic, as at the end of the derivation only the highest copy of the DP *Ken* is pronounced, and if *Ken* is the subject of MOD$_{ability}$, it will correspond to PRO, since it is in the subject position under *-rare*.